Substrate Predictor

How a P4–P4′ Substrate Window Becomes a Panel Selectivity Score

Eton iGEM 2026, Dry Lab | Tasks 7 and 8 | `spred` package

Scope. This document presents the software tool that scores an eight-residue peptide window against a panel of proteases, the methodology by which its models were built and evaluated, and the extent to which its outputs can be relied upon. It is organised into six sections, each corresponding to one stage from source data to design decision. Each section follows a consistent structure: a plain-language description of the stage; the assumptions required for the stated result, given explicitly; the governing equations, with derivations where the result is not standard; and the resulting evidence. Values obtained by direct computation, or verified against a source held in the repository, are annotated *(confirmed)*. Values that represent a working assumption, or that could not be checked independently, are annotated *(unconfirmed)*.

I. Overview: Model Structure and Operating Principle

A protease hydrolyses one peptide bond, the *scissile bond*. Residues on its amino-terminal side are numbered outward from the bond as $\mathrm{P1}, \mathrm{P2}, \ldots$ and residues on the carboxy-terminal side as $\mathrm{P1}', \mathrm{P2}', \ldots$ [1]. The enzyme surface that receives residue Pi is subsite Si, and Pi' binds Si'. Specificity is therefore positional: a protease may require proline at P3 and a large hydrophobic residue at P1′, and these preferences are largely independent of one another. The unit of data throughout this document is the eight-residue window

$$\underbrace{\mathrm{P4\ P3\ P2\ P1}}_{\text{non-primed}} \ \downarrow \ \underbrace{\mathrm{P1'\ P2'\ P3'\ P4'}}_{\text{primed}},$$

so that the scissile bond always lies between the fourth and fifth character. For example, the MMP-9 substrate SGKIPRT↓LTA measured by Kridel et al. [1] has P4P3P2P1 = IPRT and P1′P2′P3′ = LTA; its window is `IPRTLTA−`, with P4′ unknown because the peptide ends there.

The panel comprises the matrix metalloproteinases MMP-2, MMP-8, MMP-9 and MMP-14, neutrophil elastase (NE), LasB (pseudolysin of *Pseudomonas aeruginosa*), and a pooled group of digestive serine proteases (trypsin 1, chymotrypsin B and elastase-1; MEROPS S01.151–153), giving seven trainable proteases and 10,472 confirmed cleavage windows *(confirmed)*. The MMPs are zinc-dependent endopeptidases whose catalytic domain of about 170 residues carries the zinc-binding motif HEXXHXXGXXH and a conserved methionine (the "Met-turn"), together with a structural zinc and two to three calcium ions [2]. MMP-9 substrates selected by phage display carry Pro at P3 and a large hydrophobic residue at P1′, with Arg favoured at P2 and Ser/Thr at P2′; the same authors note that many MMPs recognise this Pro-X-X-Hy core and that selectivity must therefore arise from subsite interactions outside P3 and P1′ [1]. MMPs are reported to be elevated in chronic wound fluid relative to acute wound fluid, with MMP-9 the predominant elevated gelatinase, while the evidence for neutrophil elastase is mixed [3].

The tool produces, for each protease in a chosen panel, (i) a cleavage score from a logistic regression on physicochemical features, (ii) a score from a position-specific scoring matrix (PSSM), (iii) each score expressed as a percentile of random background windows, so that different proteases are comparable, and (iv) a specificity margin: the target's consensus percentile minus that of its strongest competitor. The remainder of this document is organised into six sections, each addressing one stage:

Part	Content	Section
A	Confirmed cleavage windows and negative sets	II
B	Feature representation and per-protease models	III
C	Percentile scale and panel selectivity	IV
D	Evaluation methodology and cross-validated results	V
E	Validation against independent data and structures	VI
F	Application to sensor candidate design	VII

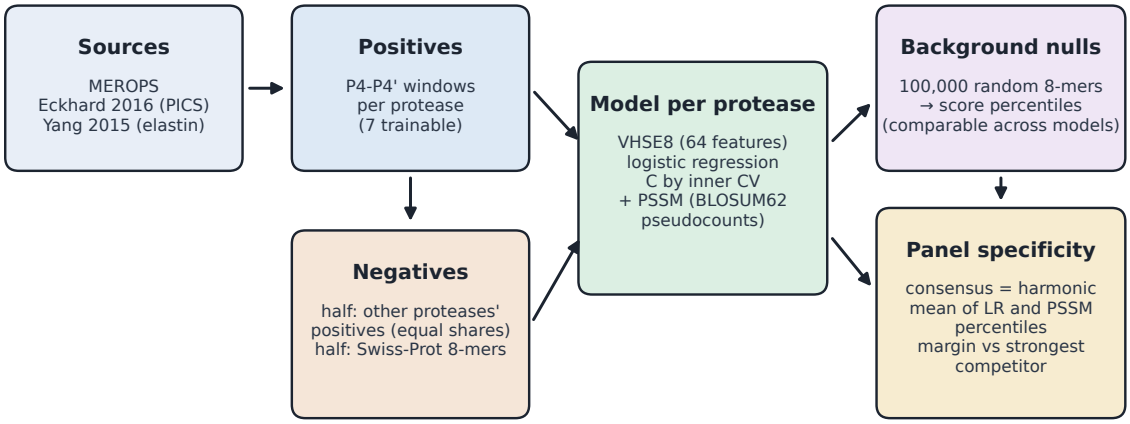


Figure 1: Pipeline from source data to panel selectivity. Positives are confirmed cleavage windows; negatives combine other proteases' windows with random background windows; each protease receives a logistic regression and a PSSM; scores are converted to percentiles of a background null and combined into a consensus and a specificity margin.

An interactive browser version of the tool accompanies this document (`substrate_predictor_explorer.html`); it embeds the trained models, the protease structures used in Part VI, and a scoring engine verified against the Python package (Part VII).

II. Part A: The Training Data

Overview

Each protease model is trained on confirmed cleavage windows. Three sources contribute: the MEROPS `Substrate_search` table [8], which lists for each recorded cleavage the eight residues around the scissile bond; the proteomic identification of cleavage sites (PICS) data of Eckhard et al. [4] for MMP-2, MMP-8, MMP-9 and MMP-14; and the elastin digestion data of Yang et al. [5] for LasB. In PICS, an MMP is incubated with peptide libraries derived from a human proteome digested with trypsin or GluC, and cleavage products are identified by LC-MS/MS at a false-discovery rate of 1%; the non-primed residues P4–P1 are recovered by locating each identified peptide in its source protein sequence. The Yang et al. study identified peptides released from bovine elastin by pseudolysin by mass spectrometry, and both termini of each released peptide are cleavage sites.

Assumptions

- **Confirmed windows are correct positives.** Windows are taken as recorded in the sources. The MEROPS table mixes physiological and synthetic substrates *(unconfirmed)*.
- **Negatives are unlabelled, not confirmed.** No window is known to be uncleaved; random background windows are assumed to be negatives. The resulting label noise is bounded in Part V but its magnitude is unknown *(unconfirmed)*.
- **Sources are pooled without correction for source bias.** PICS libraries are generated with trypsin or GluC, which constrains composition at peptide ends and may leak into primed positions of short peptides *(unconfirmed)*.
- **The LasB set is dominated by one protein.** 97 of 128 positives (76%) are windows from bovine elastin *(confirmed)*.
- **The faecal group is heterogeneous.** It pools three unrelated digestive proteases (bovine trypsin 1, human chymotrypsin B, porcine elastase-1) *(confirmed)*; its windows are capped at 5,000 of 18,251.

Sources and extraction

MEROPS windows are read from columns 5–12 of the substrate table (three-letter code for P4 to P4′), keyed by the peptidase identifier in column 2. The Eckhard windows were extracted once from the supplementary tables and archived; all peptides in the eight source sheets used carry an N-terminal label (3,536 thioacyl at +89.01 Da and 815 at +29.04 Da, the latter not defined in the source legend) *(confirmed)*. The Yang et al. Supplementary Table S1 was re-extracted directly from the supplementary PDF: all 75 peptide rows were recovered, and for each row the peptide sits at its stated position in the UniProt sequence of bovine tropoelastin (P04985) and both flanking residues match, in all 75 rows *(confirmed)*. The P4–P4′ window is then read from that sequence around both termini of each peptide. Rebuilding the MEROPS and Eckhard positive sets from the raw files reproduces the counts obtained by the earlier implementation exactly *(confirmed)*.

Protease	MEROPS	Eckhard 2016	Yang 2015	Total
MMP-2	2,318	501	0	2,819
MMP-8	96	602	0	698
MMP-9	341	442	0	783
MMP-14	129	588	0	717
Neutrophil elastase	327	0	0	327
LasB	31	0	97	128
Faecal serine proteases	5,000	0	0	5,000

The table lists positive windows per protease and source after removal of duplicate windows; a window reported by several sources is counted once, under the first.

Negative sets

For a target t with n_t positives the training set contains $N = 3n_t$ negatives. With $K - 1 = 6$ competitor proteases, a quota $q = \lfloor N/2/(K-1) \rfloor$ of windows is drawn without replacement from each competitor's positives, excluding any window that is a positive of t. The remaining negatives are random background windows whose residues are drawn independently from the UniProtKB/Swiss-Prot amino-acid composition (release 2026_03) [9], again excluding known positives. For MMP-9, $N = 2{,}349$: 195 windows from each of five competitors, 121 from LasB (its entire eligible pool), and 1,253 background windows. Construction is seeded and reproducible. Two alternatives are evaluated in Part V: background-only negatives, and the design of the earlier implementation, composition-shuffled positives plus background.

Near-duplicate structure

Sequence data of this kind contain families of near-identical windows (collagen and elastin repeats, overlapping peptides from one protein, the same site reported by two sources). If such windows are divided between training and test folds, a model can score well by recalling a neighbour. A graph is therefore defined on all training windows in which two windows are adjacent when their Hamming distance is at most 2 (at least 6 of 8 residues identical), and its connected components are termed clusters. The fraction of each positive set with a near-duplicate ranges from 2% (NE) to 28% (LasB); MMP-9 contains a single cluster of 86 windows *(confirmed)*.

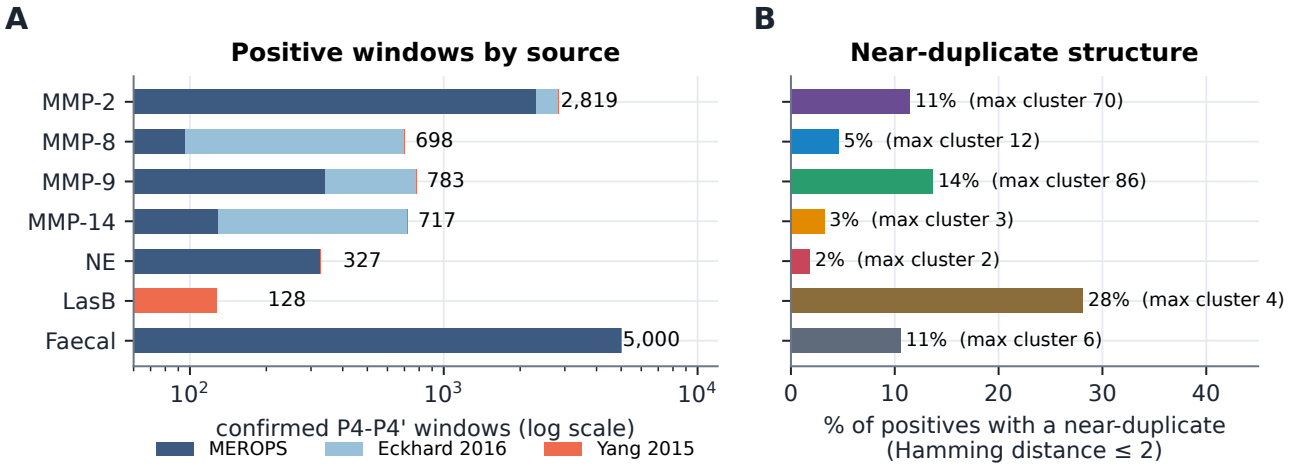


Figure 2: **A:** positive windows per protease by source (log scale). **B:** fraction of positives that have at least one near-duplicate (Hamming distance ≤ 2) and the size of the largest cluster.

Similarity between proteases

Exact window overlap between protease positive sets is small (maximum Jaccard index 0.145, between MMP-8 and MMP-9), and windows shared with the target are removed from its negatives. The position-specific preferences nevertheless overlap strongly among the MMPs: the PSSM log-odds vectors of MMP-2, MMP-8 and MMP-14 correlate at $r = 0.73$–0.82, and MMP-9 correlates at 0.56–0.63 with each. NE is essentially uncorrelated with the MMPs and correlates with the faecal group at 0.46, both being serine proteases. This structure predicts the selectivity results of Part V.

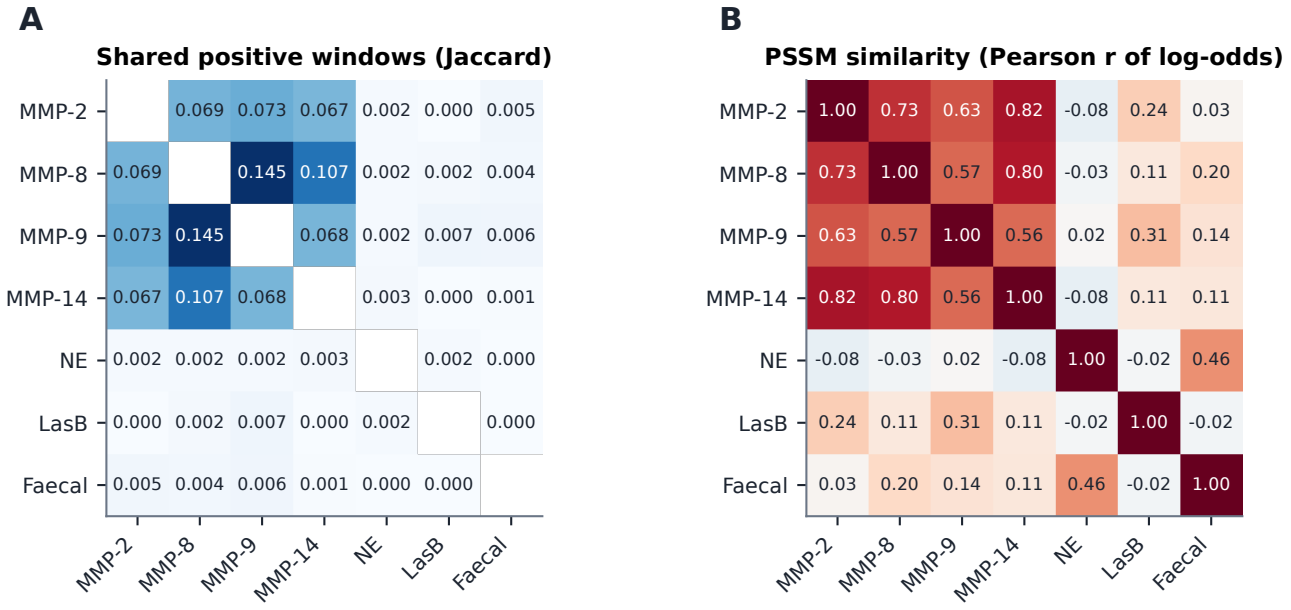


Figure 3: **A:** Jaccard index of exact positive windows. **B:** Pearson correlation of the 8×20 PSSM log-odds between protease pairs.

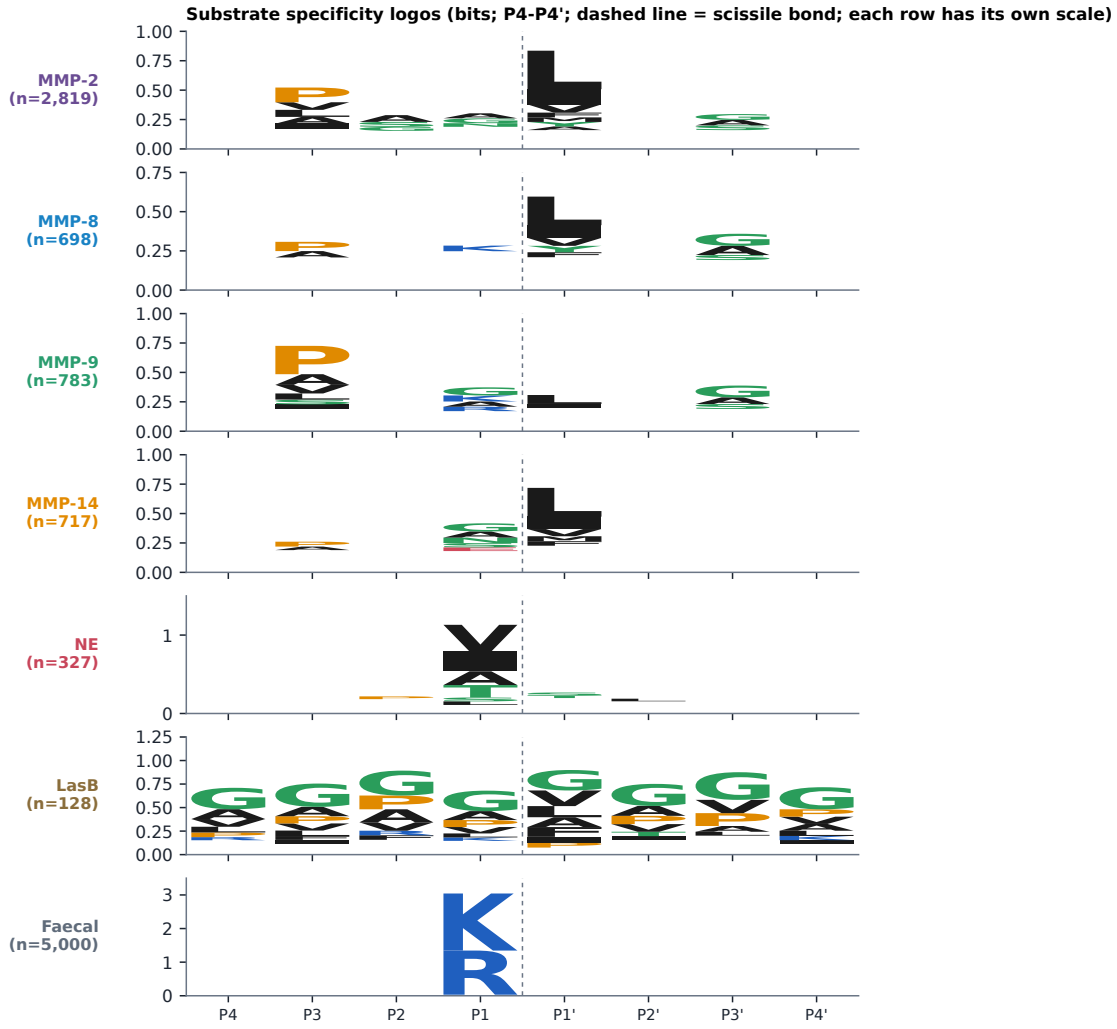


Figure 4: Specificity logos computed from each model's PSSM (letter height is probability times information content; each row has its own vertical scale, since the MMPs carry under one bit per position while the faecal group exceeds three at P1). The MMPs share Pro at P3 and a hydrophobic residue at P1′; NE prefers small aliphatic residues at P1; the faecal group is dominated by Lys/Arg at P1. The LasB logo is Gly-rich because 76% of its positives derive from elastin.

Provenance checks and corrections

Before the present models were built, each element of an earlier implementation was tested with code (audit run at repository commit `68e3dc2`; outputs retained in `results/legacy_audit/`). The findings, and the resulting action, are as follows.

Element	Finding	Action
VHSE descriptor table	35 of 160 values differed from the archived published table (several with opposite sign); glycine had been set to an all-zero reference vector, which is incorrect; descriptor labels contradicted the source definition (hydrophobic, steric, electronic); the citation named the wrong journal. Correction changed random-fold AUROC by -0.006 to $+0.012$.	Table regenerated from the archived file and unit-tested against it.
Swiss-Prot background	Two different frequency tables were in use (a residue frequency differed by up to 0.0049).	One table from release 2026_03.
Grouped cross-validation	Reported as 0.5 for every protease: the groups were the labels positive, cross and background, so every held-out group contained one class and the function returned a fixed fallback.	Replaced by cluster-aware folds.
PSSM performance	Scored on the positives used to build the PSSM. In-sample versus out-of-fold AUROC: LasB 0.977 versus 0.779, NE 0.883 versus 0.829, MMP-8 0.875 versus 0.849, MMP-9 0.816 versus 0.790.	PSSM refit and scored out-of-fold in every fold.
Selectivity index	The ratio $p_t / \max p_j$ compared probabilities from models whose mean score on random windows ranged from 0.106 to 0.328; the median index of the top 1% of MMP-9 candidates was 0.975.	Percentile scale and margin (Part IV).
LasB training data	92 positives were generated by a rule (every Ala|Ala, Val|Ala, Ala|Gly and Gly|Ala bond in a hard-coded elastin string) and attributed to a paper not held; 14 more were typed by hand; four windows tagged as MEROPS were absent from the MEROPS file. None is a measured cleavage site.	Removed. LasB uses 31 MEROPS windows and 97 from Yang et al.
Elastin table	Typed by hand from the PDF; four of 75 rows did not appear in the source.	Re-extracted from the PDF; 75 of 75 rows verified.
Kinetic table	The competing enzymes were labelled MMP-1 and MMP-2; the source gives MMP-13 and MMP-7. One measured non-substrate (C11) was omitted.	Re-extracted from the source table.
Report and code drift	Positive counts in the earlier report (MMP-14 129, MMP-8 96, MMP-9 341) no longer matched the data (717, 698, 783); a unit test asserted the incorrect glycine claim; oversampling was applied before cross-validation.	Report rewritten; test corrected; oversampling removed.
Model sprawl	Separate model files and scripts per experiment overwrote one another.	One model per protease.

Result

The corrected data layer yields seven trainable proteases with 2,819 (MMP-2), 698 (MMP-8), 783 (MMP-9), 717 (MMP-14), 327 (NE), 128 (LasB) and 5,000 (faecal group) confirmed windows. FAP is registered but has only two confirmed windows in MEROPS and cannot be trained.

III. Part B: Feature Representation and Models

Overview

Each protease receives two scorers. The first is a logistic regression on a 64-feature physicochemical encoding of the window; the second is a PSSM with BLOSUM62-derived pseudocounts. Both are additive over the eight positions, which matches the positional nature of subsite specificity, and both can therefore be decomposed exactly into one contribution per position. The physicochemical encoding uses the VHSE descriptors of Mei et al. [6]: eight principal-component score vectors derived from independent families of 18 hydrophobic, 17 steric and 15 electronic properties, of which VHSE1–2 are hydrophobic, VHSE3–4 steric and VHSE5–8 electronic. The descriptor table is archived with its source and unit-tested against it.

Assumptions

- **Position-specific effects are additive.** The effect of a residue at one position is independent of the residues elsewhere in the window. Gradient-boosted trees, which can represent interactions, do not exceed the additive models by more than 0.007 AUROC on any protease (Part V) *(confirmed)*.
- **The eight VHSE descriptors capture the relevant residue properties.** They express 64–79% of the learned per-position specificity, depending on the protease; the remainder is approximation bias *(confirmed)*.
- **The PSSM smoothing parameters are fixed.** The pseudocount weight $\beta = 5$ and the background mixing weight $\epsilon = 0.05$ were not tuned *(unconfirmed)*.
- **The** 1:3 **negative ratio is arbitrary.** It fixes the training prior but not the deployment prior, so scores are treated as rankings only *(confirmed)*.

Equations

A residue a is mapped to its VHSE vector $\varphi(a) \in \mathbb{R}^8$ and a window $w = (w_1, \ldots, w_8)$ to

$$\boldsymbol{x}(w) = \big[\varphi(w_1);\, \varphi(w_2);\, \ldots;\, \varphi(w_8)\big] \in \mathbb{R}^{64}. \tag{III.1}$$

Features are standardised with the training fold's means and standard deviations, $\boldsymbol{z} = (\boldsymbol{x} - \boldsymbol{\mu}) \oslash \boldsymbol{\sigma}$. The logistic regression minimises the class-weighted, L2-regularised loss

$$\mathcal{L}(\boldsymbol{\beta}, b) = \tfrac{1}{2}\|\boldsymbol{\beta}\|^2 + C \sum_{i=1}^{n} \omega_{y_i} \log\!\big(1 + e^{-\tilde{y}_i(\boldsymbol{\beta}^\top \boldsymbol{z}_i + b)}\big), \qquad \tilde{y}_i = 2y_i - 1, \tag{III.2}$$

with balanced class weights $\omega_c = n/(2n_c)$, that is $\omega_1 = 2$ and $\omega_0 = 2/3$ at the 1:3 ratio. Because the logit is linear in $\boldsymbol{z}$, it decomposes exactly into one term per position:

$$\operatorname{logit}(w) = b + \sum_{i=1}^{8} \kappa_i(w), \qquad \kappa_i(w) = \sum_{d=1}^{8} \beta_{id}\, z_{id}(w). \tag{III.3}$$

For the PSSM, let $f_i(a) = n_i(a)/n$ be the observed frequency of residue a at position i among n positives, W the row-normalised positive part of BLOSUM62, $q_i = f_i W$, and g the Swiss-Prot background:

$$Q_i = (1 - \epsilon)\, q_i + \epsilon\, g, \qquad p_i(a) = \frac{n\, f_i(a) + \beta\, Q_i(a)}{n + \beta}, \qquad S(w) = \sum_{i=1}^{8} \log_2 \frac{p_i(w_i)}{g(w_i)}. \tag{III.4}$$

This is a heuristic pseudocount scheme rather than the block-based estimator from which BLOSUM62 itself derives [7]. Position i carries $\mathrm{IC}_i = \sum_a p_i(a) \log_2\big(p_i(a)/g(a)\big)$ bits, which sets the letter heights in Figure 4.

Derivation

Convexity. With $m_i = \boldsymbol{\beta}^\top \boldsymbol{z}_i + b$, $p_i = \sigma(m_i)$ and $\tilde{\boldsymbol{z}}_i = [\boldsymbol{z}_i; 1]$, the gradient of Eq. (III.2) is $\nabla_{\boldsymbol{\beta}} \mathcal{L} = \boldsymbol{\beta} - C \sum_i \omega_{y_i} \tilde{y}_i\, \sigma(-\tilde{y}_i m_i)\, \boldsymbol{z}_i$, and the Hessian with respect to $(\boldsymbol{\beta}, b)$ is $\operatorname{diag}(\boldsymbol{I}, 0) + C \sum_i \omega_{y_i} p_i(1 - p_i) \tilde{\boldsymbol{z}}_i \tilde{\boldsymbol{z}}_i^\top$. For a direction $\boldsymbol{v} = (\boldsymbol{v}_\beta, v_b)$, $\boldsymbol{v}^\top H \boldsymbol{v} = \|\boldsymbol{v}_\beta\|^2 + C \sum_i \omega_{y_i} p_i(1 - p_i)(\boldsymbol{z}_i^\top \boldsymbol{v}_\beta + v_b)^2 \geq 0$, which vanishes only if $\boldsymbol{v}_\beta = \boldsymbol{0}$ and then $v_b = 0$. The Hessian is positive definite, the objective is strictly convex, and the minimiser found by L-BFGS is unique.

Meaning of the balanced weights. Equal total weight on the two classes is equivalent to fitting the posterior under a 50:50 prior. If $\rho = p_{\mathrm{bal}}/(1 - p_{\mathrm{bal}})$ is the fitted odds, the odds under the actual 1:3 training mixture are $\rho/3$, so $p_{\mathrm{mix}} = p_{\mathrm{bal}}/(3 - 2p_{\mathrm{bal}})$. Neither quantity is the probability that a random window is cleaved, which is far smaller. The logistic-regression score is therefore reported as a ranking only.

Expressive capacity of the encoding. A linear model on $\boldsymbol{x}$ gives at each position an effect $c_i^\top \varphi(w_i)$, a linear combination of the eight columns of the descriptor matrix $\Phi \in \mathbb{R}^{20 \times 8}$. Together with a constant this confines each position's twenty-dimensional effect vector to a subspace of dimension $\operatorname{rank}[\Phi, \mathbf{1}] = 9$, whereas a one-hot encoding leaves 19 free parameters per position. For a centred PSSM log-odds vector $\boldsymbol{v} \in \mathbb{R}^{20}$ and P the orthogonal projector onto $\operatorname{span}[\Phi, \mathbf{1}]$, the fraction of variance expressible by the encoding is $R^2 = 1 - \|(\boldsymbol{I} - P)\boldsymbol{v}\|^2 / \|\boldsymbol{v}\|^2$. Averaged over positions with weights $\|\boldsymbol{v}\|^2$ it lies between 64% and 79% across the proteases (table in Part V). The reduction from 19 to 8 parameters per position is the variance benefit; the unexpressed remainder is bias.

Parameters

The regularisation strength $C \in \{10^{-3}, 10^{-2}, \ldots, 10^{2}\}$ is chosen by an inner cluster-aware cross-validation (Part V). The PSSM uses $\beta = 5$ and $\epsilon = 0.05$. All random seeds are fixed at 42. Each trained model is stored as a JSON file of arrays (feature means and scales, coefficients, intercept, the 8×20 PSSM and two null quantile grids), and records the fingerprint of its training data and its cross-validated metrics.

Result

Seven models were trained. The regularisation selected by the inner loop is strong for a 64-feature model on hundreds of examples: $C = 0.01$–0.1 for every protease except the faecal group (secondary-metrics table, Part V).

IV. Part C: Percentile Scale and Panel Selectivity

Overview

Independently trained models produce scores on different scales: their logits and PSSM scores depend on the positives and negatives each was given. A selectivity measure built from raw scores, such as the ratio of the target's probability to the largest competitor probability, therefore reflects the training sets rather than the peptide. The mean logistic-regression score on random windows ranges from 0.11 (faecal group) to 0.35 (MMP-14) across the final models (Part V), which is sufficient to make such a ratio uninformative. Each score is instead expressed as a percentile of a common background distribution, and selectivity is defined on percentiles.

Assumptions

- **Random Swiss-Prot windows represent the non-substrate background.** The percentile of a window is the fraction of random windows that score below it. Some random windows may be true substrates, which lowers percentiles slightly *(unconfirmed)*.
- **The harmonic-mean consensus is a suitable combination.** It is high only when the logistic regression and the PSSM agree. It was reasoned, not tuned, and has not been validated against a selectivity ground truth *(unconfirmed)*.
- **The margin is defined relative to a chosen panel.** Adding or removing a competitor changes the margin by construction *(confirmed)*.
- **The percentile scale saturates at 10^{-3}.** The null is stored as a 1,001-point quantile grid *(confirmed)*.

Equations

For protease k let F_k^{LR} and F_k^{PSSM} be the distribution functions of its logistic-regression score and PSSM score over random background windows. The percentile scores are

$$u_k^{\mathrm{LR}}(w) = F_k^{\mathrm{LR}}\big(s_k^{\mathrm{LR}}(w)\big), \qquad u_k^{\mathrm{PSSM}}(w) = F_k^{\mathrm{PSSM}}\big(s_k^{\mathrm{PSSM}}(w)\big), \tag{IV.1}$$

and their harmonic mean is the consensus

$$u_k(w) = \frac{2\,u_k^{\mathrm{LR}}(w)\,u_k^{\mathrm{PSSM}}(w)}{u_k^{\mathrm{LR}}(w) + u_k^{\mathrm{PSSM}}(w)}. \tag{IV.2}$$

For a target t in a panel $\mathcal{P}$ the specificity margin is

$$M_t(w;\mathcal{P}) = u_t(w) - \max_{j\in\mathcal{P}\setminus\{t\}} u_j(w) \;\in [-1,1]. \tag{IV.3}$$

A margin near $+1$ indicates a window in the extreme tail for the target and near the bulk for every competitor; a margin near 0 indicates that some competitor recognises the window as strongly as the target.

Sensor cores are often shorter than eight residues around the bond, leaving some positions unknown. Both scorers are additive over positions, which permits an exact treatment. For the logit, an unknown position contributes its training mean ($z_{id} = 0$); for the PSSM it contributes the background expectation $\sum_a g(a)\log_2\big(p_i(a)/g(a)\big)$. Percentiles of such a window are taken against a null computed with the same positions masked, so that the window is compared with equally uninformed random windows.

Derivation

Comparability of percentiles. Let S have a continuous distribution function F. Then $\mathbb{P}(F(S) \le u) = \mathbb{P}(S \le F^{-1}(u)) = u$, so $F(S) \sim \mathrm{Uniform}(0,1)$. Setting F to each protease's own null distribution makes every percentile uniform on $[0,1]$ under the null: a percentile of 0.99 signifies that 1% of random windows score higher, irrespective of the raw scale of the model. For any strictly increasing g, $F_{g(S)}(g(s)) = F_S(s)$, so a percentile does not depend on the monotone form of the score (logit against probability, for instance).

Properties of the consensus. The harmonic mean $H(a,b) = 2ab/(a+b)$ satisfies $\min(a,b) \le H(a,b) \le \frac{1}{2}(a+b)$ and equals 1 only when $a = b = 1$, so the consensus is high only when both percentiles are.

Unknown positions. Let the logit be $L(\boldsymbol{z}) = b + \sum_i \boldsymbol{\beta}_i^{\top}\boldsymbol{z}_i$. If the block $\boldsymbol{z}_j$ of an unknown position is random with mean $\mathbf{0}$ and independent of the rest, then $\mathbb{E}[L] = b + \sum_{i\neq j}\boldsymbol{\beta}_i^{\top}\boldsymbol{z}_i$, which is L with $\boldsymbol{z}_j = \mathbf{0}$. The identity is exact for the logit and does not hold for $\sigma(L)$, which is why the imputation is applied to the logit. For the PSSM the expected contribution of position j under the background is

$$\sum_a g(a)\log_2\frac{p_j(a)}{g(a)} = -D_{\mathrm{KL}}(g\,\|\,p_j) \le 0, \tag{IV.4}$$

so an unknown position lowers the score by an amount that grows with the information carried by that position.

Parameters

The null for full windows uses 100,000 random windows, stored as a 1,001-point quantile grid and interpolated linearly. The null for gapped windows uses 20,000 random windows per gap pattern. Random windows are drawn from the Swiss-Prot composition [9].

Result

Percentile scores place every protease on one scale and permit a panel-level margin. Under the null the percentile is uniform by construction; on held-out positives it concentrates near the top of the scale for every protease (Part V).

V. Part D: Evaluation Methodology and Results

Overview

Performance is estimated by cross-validation in which folds are formed over clusters of near-duplicate windows, regularisation is chosen inside each fold, and all intervals are cluster-bootstrap intervals. Model choices are tested against baselines on identical folds with paired significance tests. Two quantities are evaluated: *detection*, whether held-out confirmed sites for a protease are ranked above unlabelled windows, and *discrimination*, whether they are ranked above confirmed sites for a different protease. The second is the property a multiplexed sensor requires, and it is the harder one.

Assumptions

- **Label noise can only lower AUROC.** If a fraction π of the negatives are hidden positives, the measured AUROC is $(1-\pi)\,\mathrm{AUROC}_{\mathrm{true}} + \frac{1}{2}\pi$ (derivation below) *(confirmed).*
- **Clusters at Hamming distance 2 capture the dependence between windows.** Dependence at greater distance, for example homologous proteins, is not modelled *(unconfirmed).*
- **Cross-validation shares the data-generating process of the training data.** It cannot detect bias common to all sources; independent validation is reported in Part VI *(confirmed).*
- **No multiplicity correction is applied.** Of 35 paired model comparisons, 28 are tabulated; isolated borderline results should be treated with caution *(confirmed).*

Design

Cluster-aware folds. Clusters are the connected components of the Hamming-distance-≤ 2 graph over all training windows. They are found exactly by a pigeonhole argument: partition the eight positions into blocks $\{1,2,3\}$, $\{4,5,6\}$ and $\{7,8\}$; two windows differing at no more than two positions must agree exactly on at least one block, so only windows that share a block need be compared, and connected components follow by union–find. Folds are drawn by stratified group 5-fold cross-validation over clusters, so that no cluster straddles a train/test boundary and each fold preserves the class ratio. All metrics are computed on out-of-fold predictions pooled across folds.

Nested regularisation. Within each outer training set, C is chosen from $\{10^{-3},\dots,10^{2}\}$ by an inner 3-fold cluster-aware cross-validation on mean AUROC, and only then is the model refit and applied to the outer test fold. The PSSM is rebuilt from each training fold's positives and scored out-of-fold. The background null used for percentiles is rebuilt per fold from that fold's models (5,000 random windows), so the cross-validated consensus uses no information from its test fold.

Metrics. AUROC is the probability that a random positive outranks a random negative; it is threshold-free, invariant to any increasing rescaling of the score, and unaffected by the class ratio. PR-AUC is measured against a prior of 0.25. MCC uses the logistic-regression score at the fixed threshold 0.5 and is only indicative under balanced class weighting. The Brier score and expected calibration error (ECE) describe agreement between score and outcome under the training prior. Intervals are 95% cluster-bootstrap intervals (500 replicates); for model comparisons the same resampled clusters are used for both models (paired cluster bootstrap, 400 replicates).

Derivation

AUROC as a Mann–Whitney probability. Let S^+ and S^- be independent scores of a random positive and a random negative, with survival functions $\bar{F}_1$ and $\bar{F}_0$ and density f_0. The ROC curve traces $(\bar{F}_0(\tau), \bar{F}_1(\tau))$ as τ decreases, so

$$\mathrm{AUROC} = \int \mathrm{TPR}\, d\mathrm{FPR} = \int_{-\infty}^{\infty} \bar{F}_1(\tau)\, f_0(\tau)\, d\tau = \mathbb{P}(S^+ > S^-), \tag{V.1}$$

with a further $\frac{1}{2}\mathbb{P}(S^+ = S^-)$ in the presence of ties. The empirical estimator is the Mann–Whitney statistic $U/(n_+ n_-)$.

Label contamination. Suppose a negative is, with probability π, a hidden positive drawn from the labelled-positive distribution and otherwise a true negative. For a fixed continuous scorer,

$$\mathrm{AUROC}_{\mathrm{measured}} = (1-\pi)\mathbb{P}(S^+ > S^-_{\mathrm{true}}) + \pi\,\mathbb{P}(S^+ > S^{+\prime}) = (1-\pi)\,\mathrm{AUROC}_{\mathrm{true}} + \tfrac{1}{2}\pi, \tag{V.2}$$

since two independent draws from one continuous distribution are ordered either way with probability $\frac{1}{2}$. Contamination can only lower AUROC, and the measured value is at most $1-\pi/2$ even for a perfect scorer.

Results: detection performance

Cluster-aware, out-of-fold AUROC for the logistic regression, the PSSM and their consensus (Eq. IV.2) is given below (bold: highest of the three). Discrimination is well above chance for every protease, from 0.772 for MMP-9 to 0.953 for the faecal group. The consensus is at or near the best single scorer for every protease; for MMP-2 and MMP-8 the PSSM alone is slightly higher, with overlapping intervals.

Protease	Positives	Positive clusters	LR AUROC	PSSM AUROC	Consensus AUROC
MMP-2	2,819	2,608	0.854 [0.844, 0.863]	**0.874 [0.863, 0.883]**	0.869 [0.859, 0.878]
MMP-8	698	677	0.755 [0.733, 0.774]	**0.784 [0.762, 0.800]**	0.783 [0.763, 0.800]
MMP-9	783	687	0.760 [0.723, 0.791]	0.761 [0.723, 0.798]	**0.772 [0.737, 0.803]**
MMP-14	717	705	0.772 [0.753, 0.790]	0.785 [0.766, 0.806]	**0.791 [0.771, 0.810]**
Neutrophil elastase	327	324	0.895 [0.875, 0.913]	0.885 [0.861, 0.906]	**0.901 [0.880, 0.919]**
LasB	128	106	0.877 [0.835, 0.917]	0.881 [0.837, 0.916]	**0.887 [0.844, 0.923]**
Faecal serine proteases	5,000	4,715	0.951 [0.948, 0.954]	0.953 [0.950, 0.956]	**0.953 [0.950, 0.956]**

Secondary metrics for the logistic regression follow. PR-AUC is measured against a prior of 0.25; MCC at threshold 0.5; ECE under the training prior; the selected C is listed for each outer fold.

Protease	PR-AUC	MCC	Brier	ECE	C chosen per outer fold
MMP-2	0.649	0.521	0.156	0.153	0.1, 0.1, 0.1, 0.1, 0.1
MMP-8	0.474	0.346	0.202	0.199	0.01, 0.01, 0.01, 0.01, 0.01
MMP-9	0.535	0.335	0.196	0.196	0.01, 0.01, 0.01, 0.01, 0.01
MMP-14	0.516	0.370	0.195	0.191	0.1, 0.01, 0.01, 0.1, 0.01
Neutrophil elastase	0.751	0.600	0.127	0.114	0.1, 0.1, 0.1, 0.1, 0.1
LasB	0.783	0.602	0.125	0.122	0.01, 0.1, 0.1, 0.1, 0.01
Faecal serine proteases	0.800	0.806	0.071	0.055	1.0, 0.1, 1.0, 0.1, 0.1

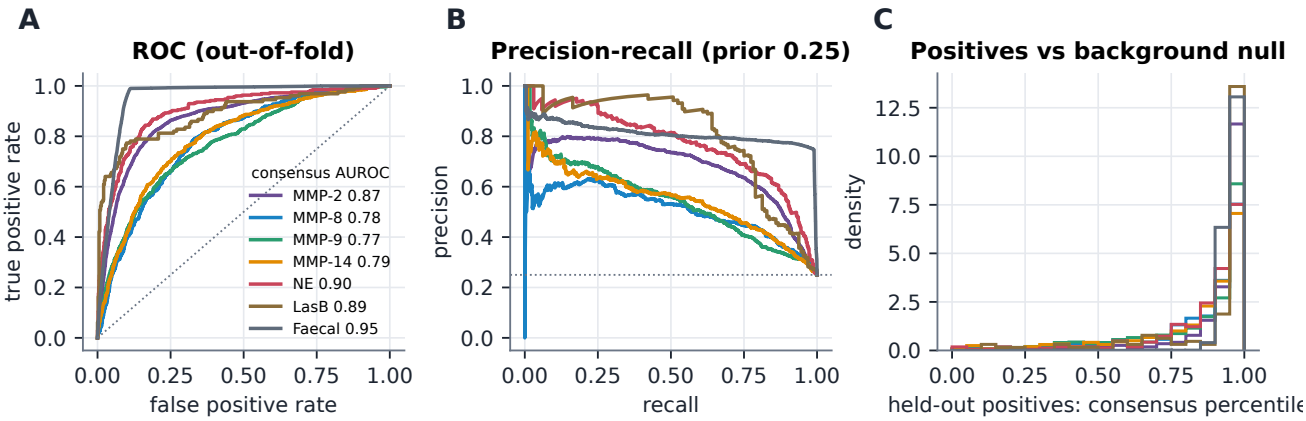


Figure 5: Out-of-fold performance of the consensus score. **A:** ROC curves. **B:** precision-recall curves. **C:** distribution of the consensus percentile of held-out positives, which concentrates near the top of the background distribution for every protease.

Results: model and representation comparison

Seven models were compared on identical folds: a constant prior, a composition-only logistic regression, one-hot and VHSE logistic regressions, the PSSM, and gradient-boosted trees on both encodings. Three conclusions follow.

- **Position matters and composition mostly does not.** For six of seven proteases composition alone gives 0.60–0.74 AUROC and the VHSE model exceeds it significantly; label-shuffled controls give 0.49–0.53. LasB is the exception: composition alone reaches 0.886, statistically indistinguishable from the full model. The LasB model therefore detects largely elastin-like composition, as expected from a positive set that is 76% elastin *(confirmed)*.
- **No strong position–position interaction is evident.** Gradient-boosted trees reach within 0.013 AUROC of the better of the one-hot regression and the PSSM on every protease and exceed it only on NE, by 0.007 *(confirmed)*.
- **The representation trade-off is real.** On data-rich proteases (MMP-2, MMP-8) the one-hot regression and the PSSM exceed the VHSE model significantly, by 0.017 to 0.034 AUROC, consistent with the bias measured by the subspace coverage below. On NE and LasB the VHSE model exceeds the one-hot regression significantly (+0.021 and +0.033). The learning curves show the same crossover: VHSE leads at small training sizes and is overtaken as data grow *(confirmed)*.

The PSSM has no fitted hyperparameter and lies at or within the interval of the best model for every protease, which is the reason the shipped score combines it with the logistic regression.

Protease	Prior	Composition	One-hot LR	VHSE LR	PSSM	One-hot GBM	VHSE GBM
MMP-2	0.500	0.664	0.871	0.854	**0.874**	0.869	0.872
MMP-8	0.500	0.596	0.777	0.750	**0.784**	0.780	0.783
MMP-9	0.500	0.639	0.756	0.754	**0.761**	0.752	0.755
MMP-14	0.500	0.633	0.772	0.771	**0.785**	0.773	0.779
Neutrophil elastase	0.500	0.676	0.866	0.887	0.885	0.892	**0.894**
LasB	0.500	0.886	0.845	0.877	0.881	0.878	**0.889**
Faecal serine proteases	0.500	0.742	0.952	0.951	**0.953**	0.950	0.951

The table gives cross-validated AUROC by model (bold: best); the prior gives 0.5. The next table gives the AUROC of each alternative minus that of the VHSE logistic regression under a paired cluster bootstrap (asterisk: 95% interval excludes zero).

Protease	One-hot LR	PSSM	One-hot GBM	VHSE GBM
MMP-2	+0.017*	+0.020*	+0.015*	+0.018*
MMP-8	+0.027*	+0.034*	+0.029*	+0.034*
MMP-9	+0.003	+0.008	-0.002	+0.001
MMP-14	+0.000	+0.014*	+0.002	+0.009
Neutrophil elastase	-0.021*	-0.002	+0.005	+0.007
LasB	-0.033*	+0.003	+0.000	+0.011
Faecal serine proteases	+0.000	+0.001	-0.001	-0.000

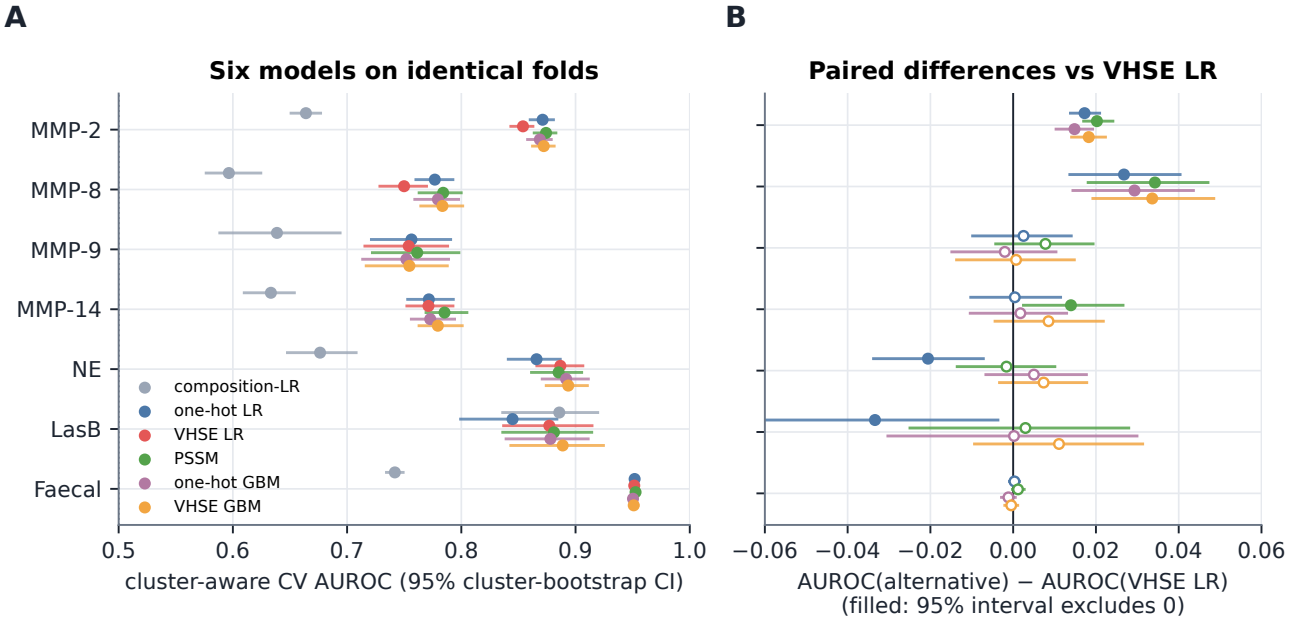


Figure 6: **A:** cluster-aware AUROC of six models on identical folds with 95% cluster-bootstrap intervals. **B:** paired difference (alternative minus VHSE logistic regression); filled markers have intervals excluding zero. The composition-only model lies 0.12–0.21 below for every protease except LasB and is omitted from panel B for scale.

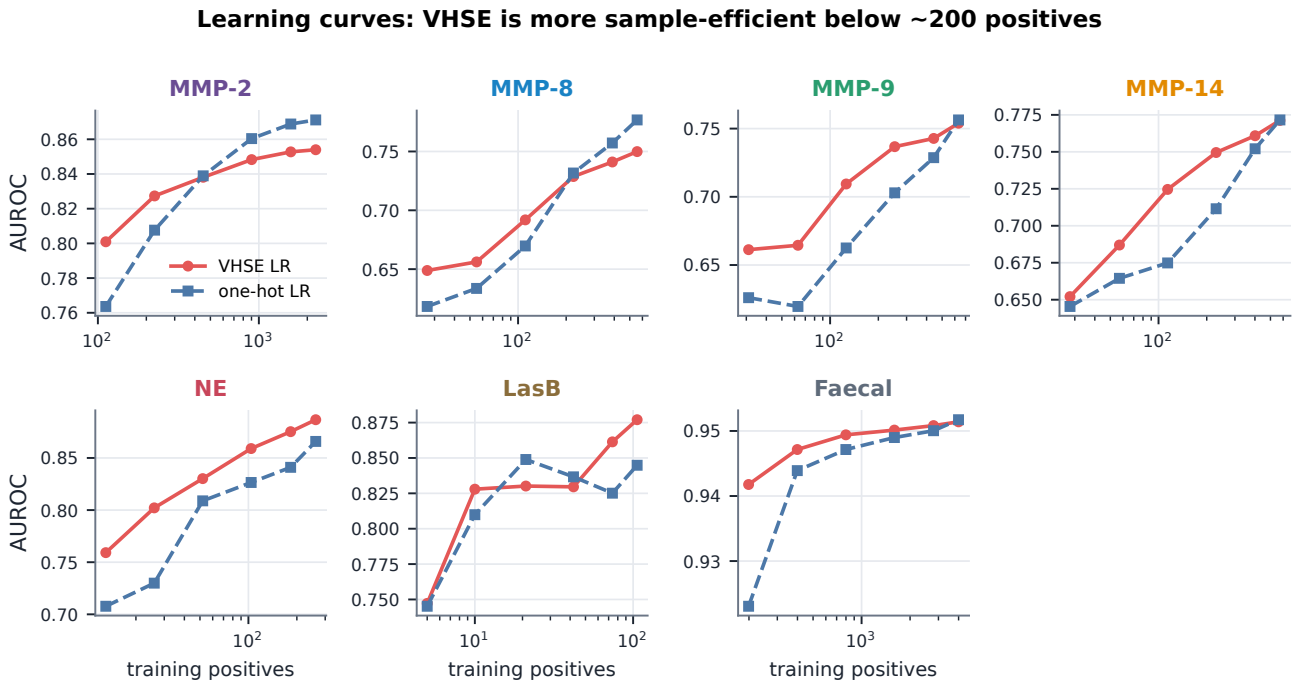


Figure 7: Learning curves: AUROC as a function of the number of training positives for the VHSE and one-hot logistic regressions.

Results: training against other proteases

Three negative-set designs were compared on a common held-out test set within identical folds: composition-shuffled positives plus background (the earlier design), background only, and the panel design of Part II. Each was scored on held-out target positives against background windows and against held-out positives of the other proteases. The panel design costs 0.01–0.02 AUROC against background and gains against competitors on every protease, most where it matters: for LasB from 0.80 (background only) and 0.68 (shuffled) to 0.86, for MMP-14 from 0.65 to 0.73, and for MMP-2 from 0.76 to 0.79 *(confirmed)*.

	Shuffled + background		Background only		Panel (ours)	
Protease	vs bkg	vs comp.	vs bkg	vs comp.	vs bkg	vs comp.
MMP-2	0.881	0.756	0.885	0.762	0.882	0.787
MMP-8	0.834	0.654	0.828	0.651	0.810	0.683
MMP-9	0.819	0.660	0.832	0.667	0.816	0.683
MMP-14	0.831	0.660	0.833	0.649	0.811	0.727
Neutrophil elastase	0.850	0.910	0.852	0.900	0.848	0.926
LasB	0.801	0.676	0.889	0.798	0.891	0.863
Faecal serine proteases	0.954	0.939	0.954	0.938	0.953	0.944

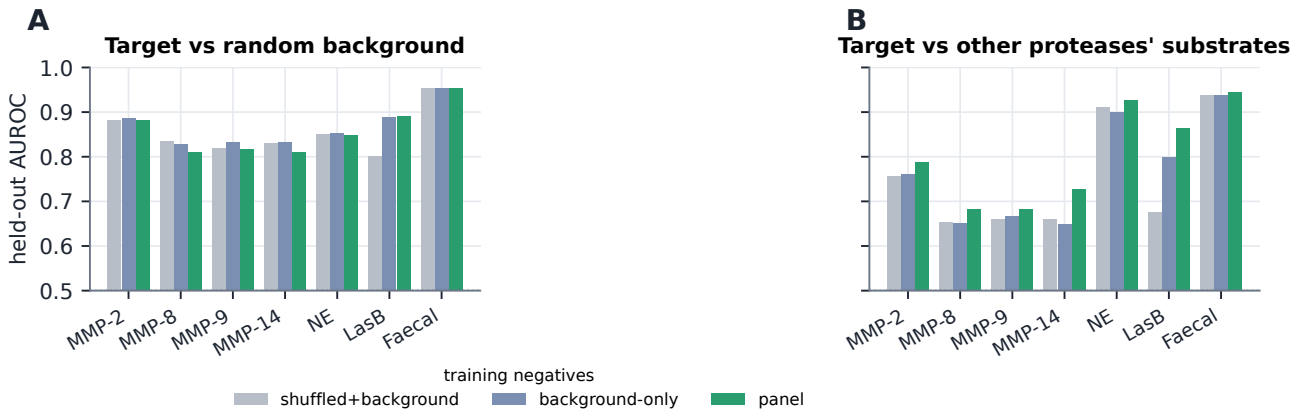


Figure 8: Effect of the training negatives on detection (**A**) and cross-protease discrimination (**B**).

Results: regularisation, leakage and controls

Choosing C by inner cross-validation raises or matches the $C = 1$ default, selecting $C = 0.01$–0.1 for every protease except the faecal group. Random and cluster-aware folds differ by at most 0.007 AUROC, so near-duplicate leakage is not a material source of optimism in these data, although the folds guard against it. The table gives regularisation (AUROC at $C = 1$ against nested selection), fold construction (random against clustered, fixed $C = 0.1$), the label-shuffle control (mean of five permutations), and the variance-weighted share of the learned PSSM log-odds that lies inside the VHSE subspace.

	Regularisation		Fold construction		Label	VHSE span
Protease	$C = 1$	nested	random	clustered	shuffle	coverage
MMP-2	0.854	0.854	0.855	0.854	0.505	79%
MMP-8	0.750	0.755	0.752	0.752	0.493	64%
MMP-9	0.754	0.760	0.755	0.756	0.493	64%
MMP-14	0.771	0.772	0.771	0.773	0.509	79%
Neutrophil elastase	0.887	0.895	0.897	0.895	0.495	65%
LasB	0.877	0.877	0.886	0.879	0.526	65%
Faecal serine proteases	0.951	0.951	0.951	0.951	0.494	77%

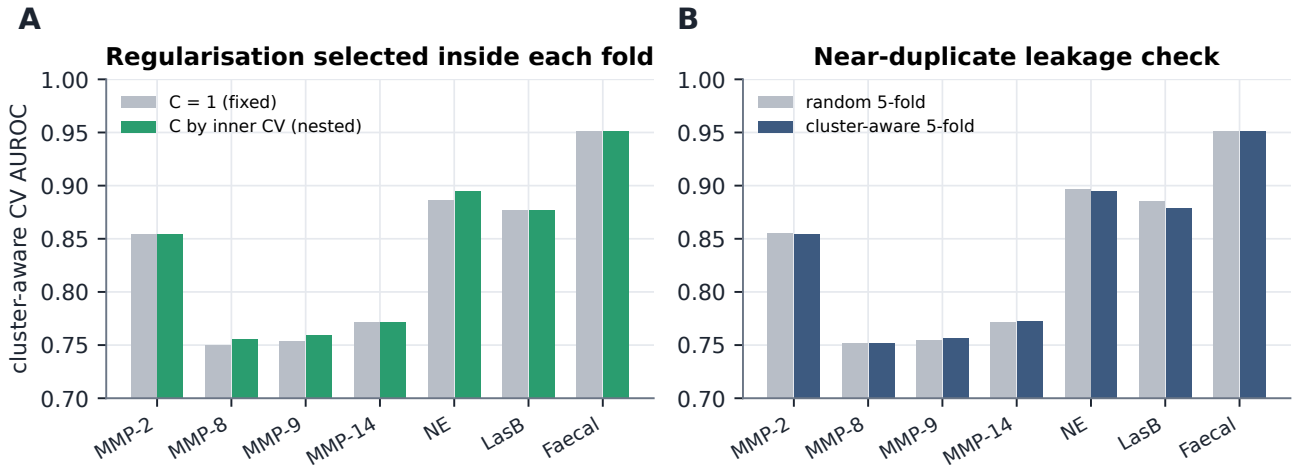


Figure 9: **A:** nested regularisation selection against fixed $C = 1$. **B:** random against cluster-aware folds.

Results: calibration and scale

Reliability curves of the logistic-regression score lie below the diagonal: with balanced class weights the score is a posterior under a 50:50 prior, whereas the training mixture has a positive prior of 0.25 and real sequences a far lower one. Expected calibration error is 0.06–0.20. Base rates also differ by model, the mean score on random windows ranging from 0.11 (faecal group) to 0.35 (MMP-14), which motivates the percentile scale of Part IV *(confirmed)*.

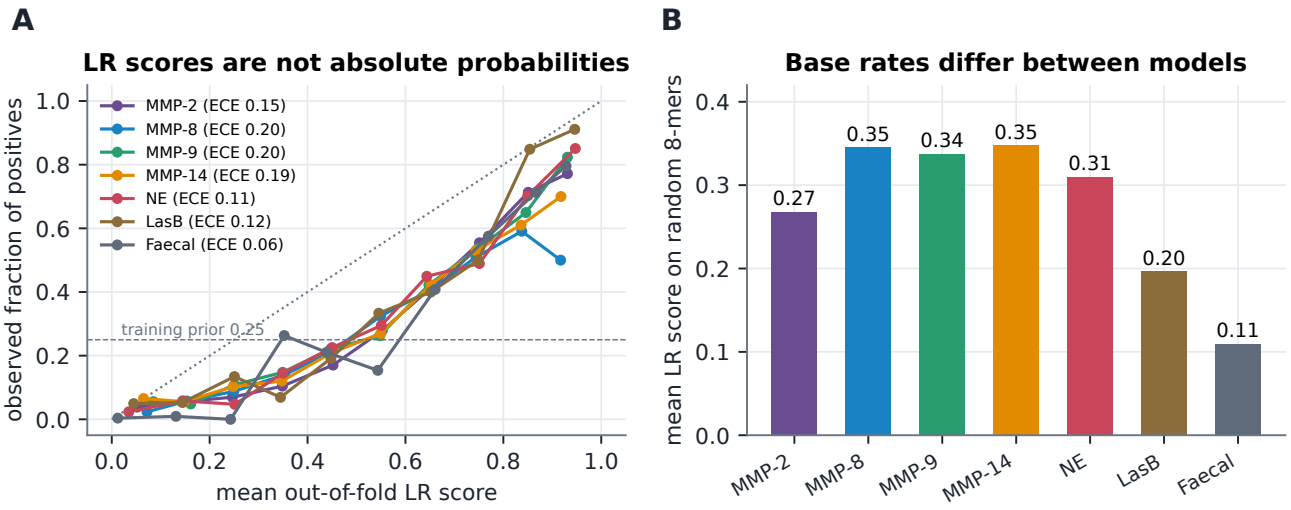


Figure 10: **A:** reliability curves of the out-of-fold logistic-regression score. **B:** mean logistic-regression score on random background windows for each final model.

Results: selectivity between proteases

The next table gives, for each target model, the held-out AUROC for separating the target's substrates from a competitor's substrates (the competitor's windows were training negatives, but each is scored out-of-fold). The NE and faecal models separate their substrates from every competitor's at 0.90–0.98. The MMP models separate theirs from NE's at 0.89–0.93, but from LasB's and the faecal group's at only 0.63–0.89. Within the MMP family discrimination is modest (0.53–0.72), lowest for MMP-14 against MMP-2 (0.53) and MMP-8 against MMP-2 (0.57) *(confirmed)*.

Target model	MMP-2	MMP-8	MMP-9	MMP-14	NE	LasB	Faecal
MMP-2	–	0.72	0.68	0.69	0.93	0.79	0.89
MMP-8	0.57	–	0.58	0.64	0.91	0.82	0.63
MMP-9	0.59	0.65	–	0.70	0.90	0.63	0.63
MMP-14	0.53	0.62	0.70	–	0.89	0.80	0.84
Neutrophil elastase	0.93	0.93	0.93	0.94	–	0.92	0.96
LasB	0.84	0.90	0.81	0.84	0.87	–	0.91
Faecal	0.96	0.90	0.91	0.96	0.98	0.97	–

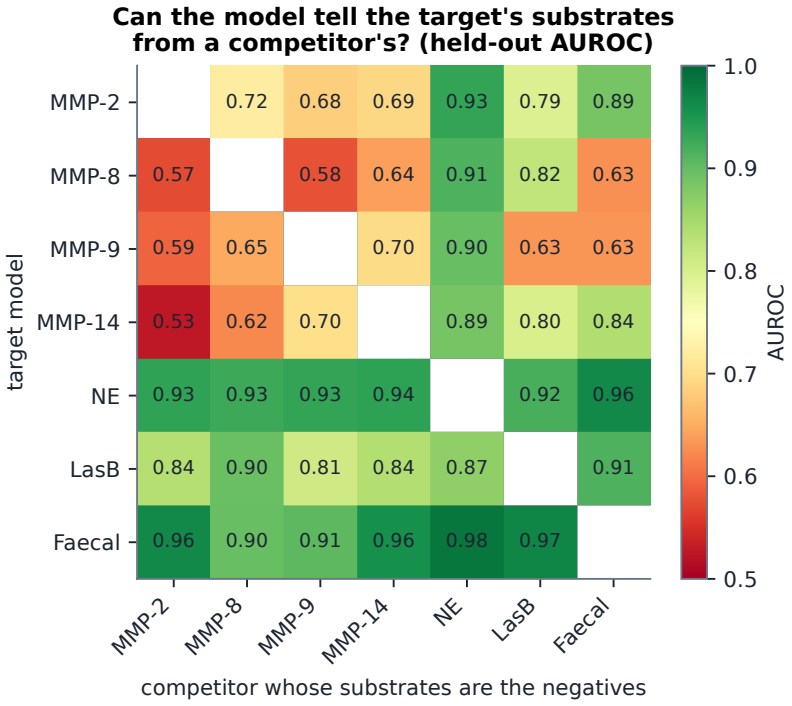


Figure 11: Cross-protease discrimination matrix (data of the preceding table).

A low competitor AUROC does not by itself indicate a poor model. By Eq. (V.2), if a fraction π of a competitor's substrates are also cleaved by the target, the measured AUROC is at most $1-\pi/2$ for a perfect model. Exact-window overlap between protease positive sets is small, but the MMPs share their dominant preferences [1], so many competitor windows are plausibly cleaved by the target as well; the ceiling for MMP-against-MMP AUROC is therefore below 1 by an amount that sequence data alone cannot measure *(unconfirmed)*. The practical conclusion is unchanged: from a P4–P4′ window alone the MMPs cannot be reliably distinguished, and a sensor design should not depend on it.

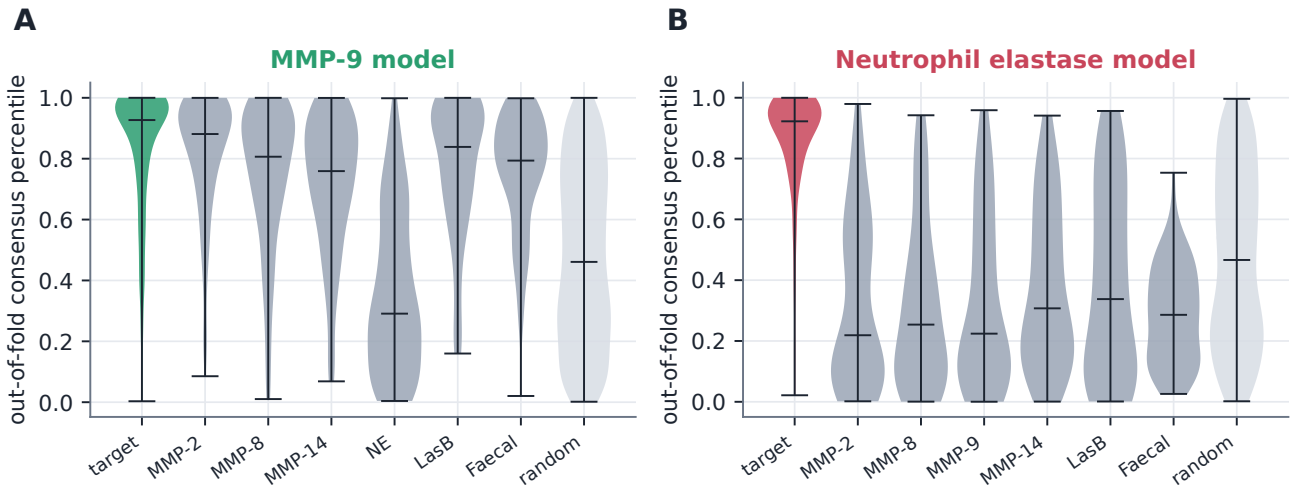


Figure 12: Out-of-fold consensus percentiles of held-out positives (target) and of each competitor's substrates and random windows for the MMP-9 model (**A**) and the NE model (**B**). The MMP-9 model places MMP-2 and MMP-8 substrates almost as high as its own; the NE model separates them sharply.

Results: what the models learned

Coefficient stability by cluster bootstrap identifies which learned effects are reliable: between 17 (MMP-14) and 35 (MMP-2) of the 64 standardised coefficients per model have 95% intervals excluding zero, so the position-level

pattern is more trustworthy than any single descriptor weight. Total contribution is concentrated at P3 and P1′ for the MMPs, at P1 for NE and the faecal group, and is spread across positions for LasB; the electronic family, which accounts for four of the eight descriptors, carries the largest share for every protease.

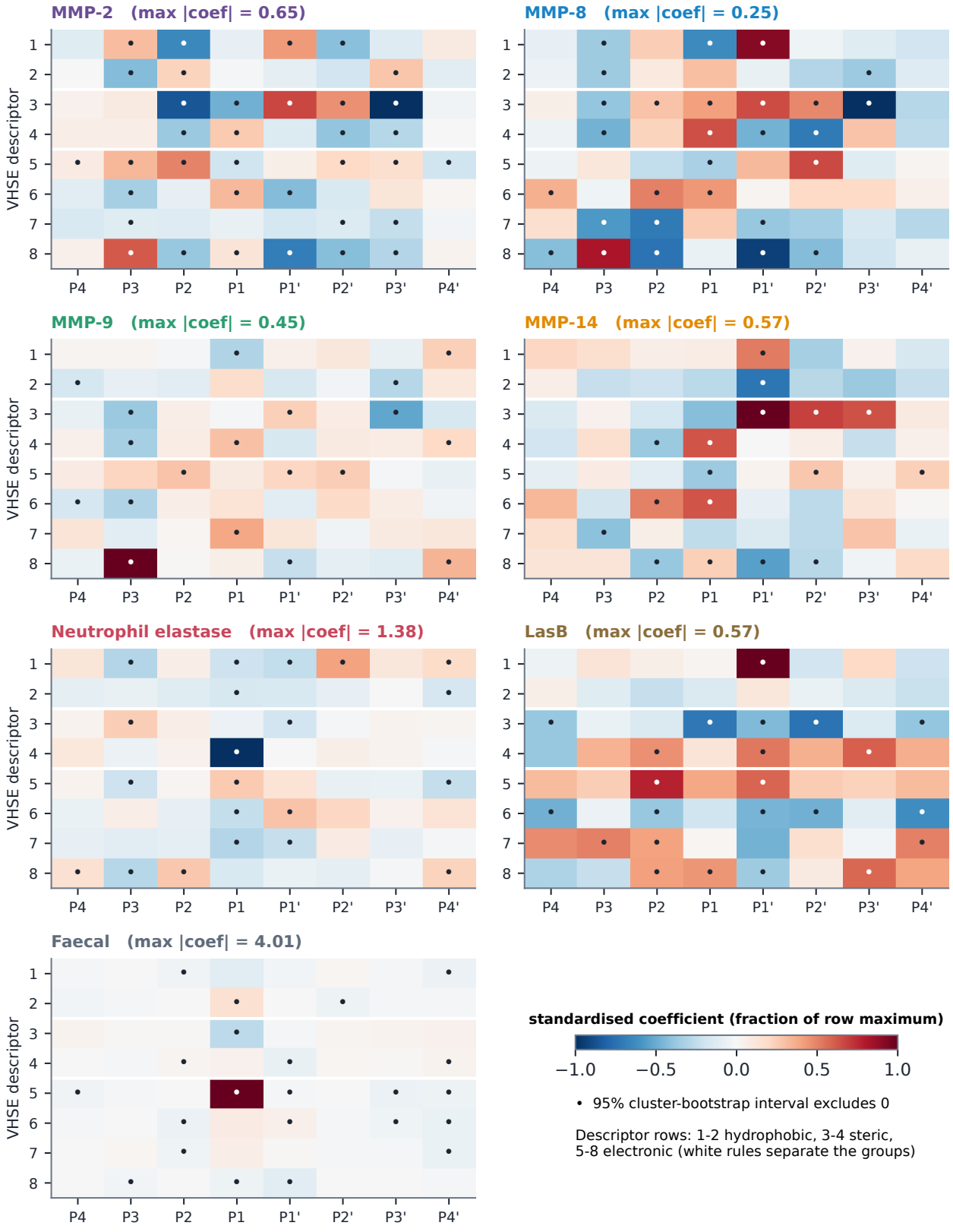


Figure 13: Standardised logistic-regression coefficients by position and VHSE descriptor. Dots mark coefficients whose 95% cluster-bootstrap interval excludes zero. Colours are scaled to each protease's largest coefficient.

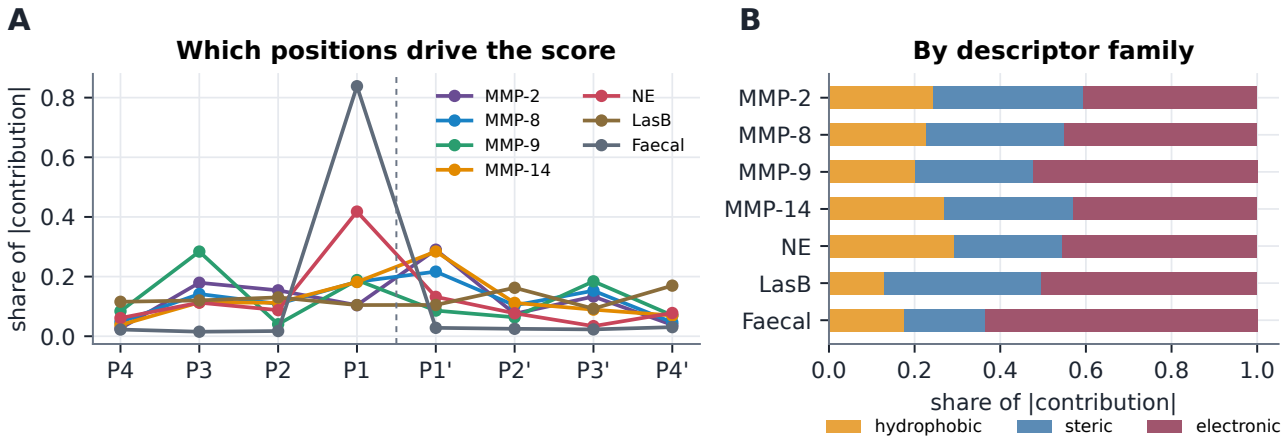


Figure 14: **A:** share of mean absolute logit contribution by position, computed on training positives. **B:** share by descriptor family.

VI. Part E: Validation Against Independent Data and Structures

Overview

Cross-validation shows that the models generalise within their data-generating process. Three comparisons that do not share that process are reported: whether the model recovers published specificity that it was not shown, whether the active-site structure of the enzymes agrees with what it learned, and whether its scores rank measured kinetics.

Assumptions

- **The phage-display consensus of Kridel et al. is independent of the training data.** It derives from a different experiment and is not in MEROPS or PICS [1] *(unconfirmed)*.
- **Kinetic values are transcribed from Table II of the source.** All twelve rows were extracted programmatically and checked against the paper [1] *(confirmed)*.
- **Peptides shorter than eight positions around the bond are scored as gapped windows.** The peptides are N-acetylated and C-amidated, so P4′ is absent for most of them *(confirmed)*.
- **The validation set is small and pre-selected.** Eleven peptides, all selected for MMP-9 activity, with no MMP-13 or MMP-7 model available to test the reported selectivity *(confirmed)*.

Recovery of the MMP-9 consensus

The consensus of Kridel et al., Pro-Arg-Ser/Thr↓Hy-Ser/Thr, is overlaid on the learned MMP-9 PSSM below. Pro at P3 is the top-ranked residue of twenty (+2.80 bits), Leu and Ile are first and second at P1′ (+1.36 and +0.85 bits), and Arg at P2 ranks 4 of 20 (+0.35 bits) *(confirmed)*. These are the two features that Kridel et al. associate with specific enzyme pockets: an S3 subsite that accommodates the bend introduced by Pro, and the deep hydrophobic S1′ pocket.

Ser/Thr at P1 and P2′ is not reproduced (Thr ranks 16 of 20 at P1, −1.14 bits; Ser ranks 18 at P2′). This is consistent with the report of Kridel et al. that about two thirds of the P2′ Thr residues in the selected clones are encoded by the phage vector rather than selected. At P1 the top residues are Gly, Lys and Arg (+1.39, +1.25, +0.87 bits); Gly at P1 is the pattern of the natural collagen substrates in Table III of the same source, and Arg at P1 is favoured in their group III substrates. The learned specificity is that of natural and proteome-derived sites, which differs from a phage-optimised consensus at the positions where that difference is expected.

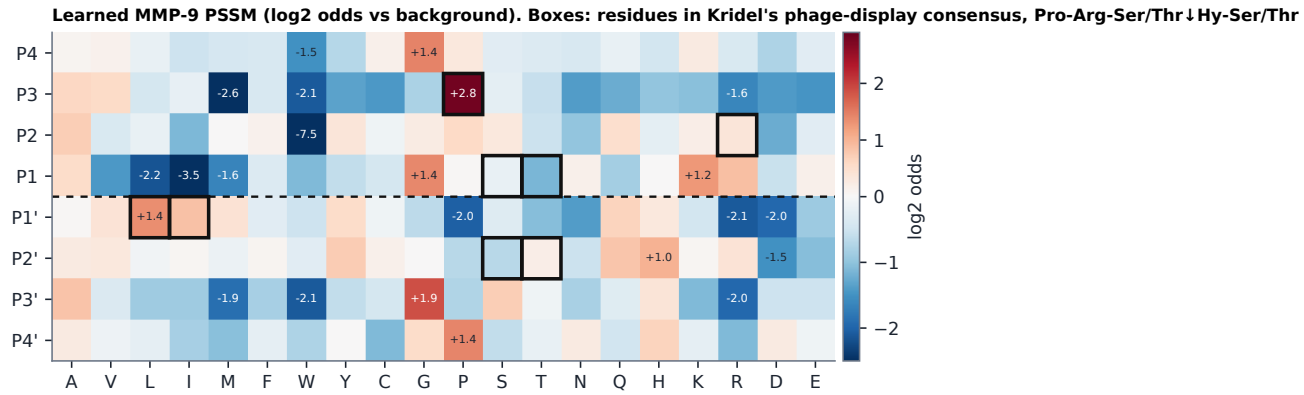


Figure 15: Learned MMP-9 PSSM ($\log_2$ odds against background; cells with $|\cdot| > 1$ bit are annotated). Boxes mark the residues of the Kridel et al. phage-display consensus. Pro at P3 and Leu/Ile at P1′ are recovered; Ser/Thr at P1 and P2′ are not.

Learned specificity and the enzyme structures

One experimental structure per protease was prepared (table below), and active-site facts were derived from the coordinates. In MMP-9, MMP-2 and MMP-8 the sequence around the catalytic zinc reads H-E-F-G-H-A-L-G-L-D/E-H, the HEXXHXXGXXH motif of Nagase and Woessner [2] *(confirmed)*; the zinc is coordinated by the two histidines of HEXXH and the third histidine of the motif (His401, His405 and His411 in MMP-9), and the hydroxamate oxygen of the bound MMP-9 inhibitor lies 2.07 Å from it *(confirmed)*. In LasB the zinc ligands are His140, His144 and Glu164; in neutrophil elastase the catalytic triad is His57, Asp102 and Ser195 *(confirmed)*. Residues lining the ligand-occupied pocket of MMP-9 include Asp185-Gly186-Leu187, the Asp-Gly-Leu segment that Kridel et al. assign to the S2′ subsite (their numbering differs), and Met422. In NE they include Val190, Ala213 and Val216, small aliphatic side chains, and the top residues of the NE model at P1 are Val, Ile and Thr.

Protease	PDB	Resolution (Å)	Residues	Catalytic site residues	Pocket residues
MMP-2	1QIB	2.8	160	HIS201, HIS205, HIS211	23
MMP-8	1JAN	2.5	164	HIS197, HIS201, HIS207	21
MMP-9	1GKC	2.3	159	HIS401, HIS405, HIS411	20
Neutrophil elastase	1HNE	1.84	210	ASP102, HIS57, SER195	22
LasB	1EZM	1.5	298	GLU164, HIS140, HIS144	27

The table lists the structures used. Catalytic-site residues are those coordinating the catalytic zinc (MMPs, LasB) or forming the catalytic triad (NE); pocket residues lie within 6 Å of the bound ligand, or within 9 Å of the catalytic zinc when no ligand is present. MMP-9 is taken from an active catalytic-domain entry rather than the pro-form, whose prodomain occupies the cleft [10].

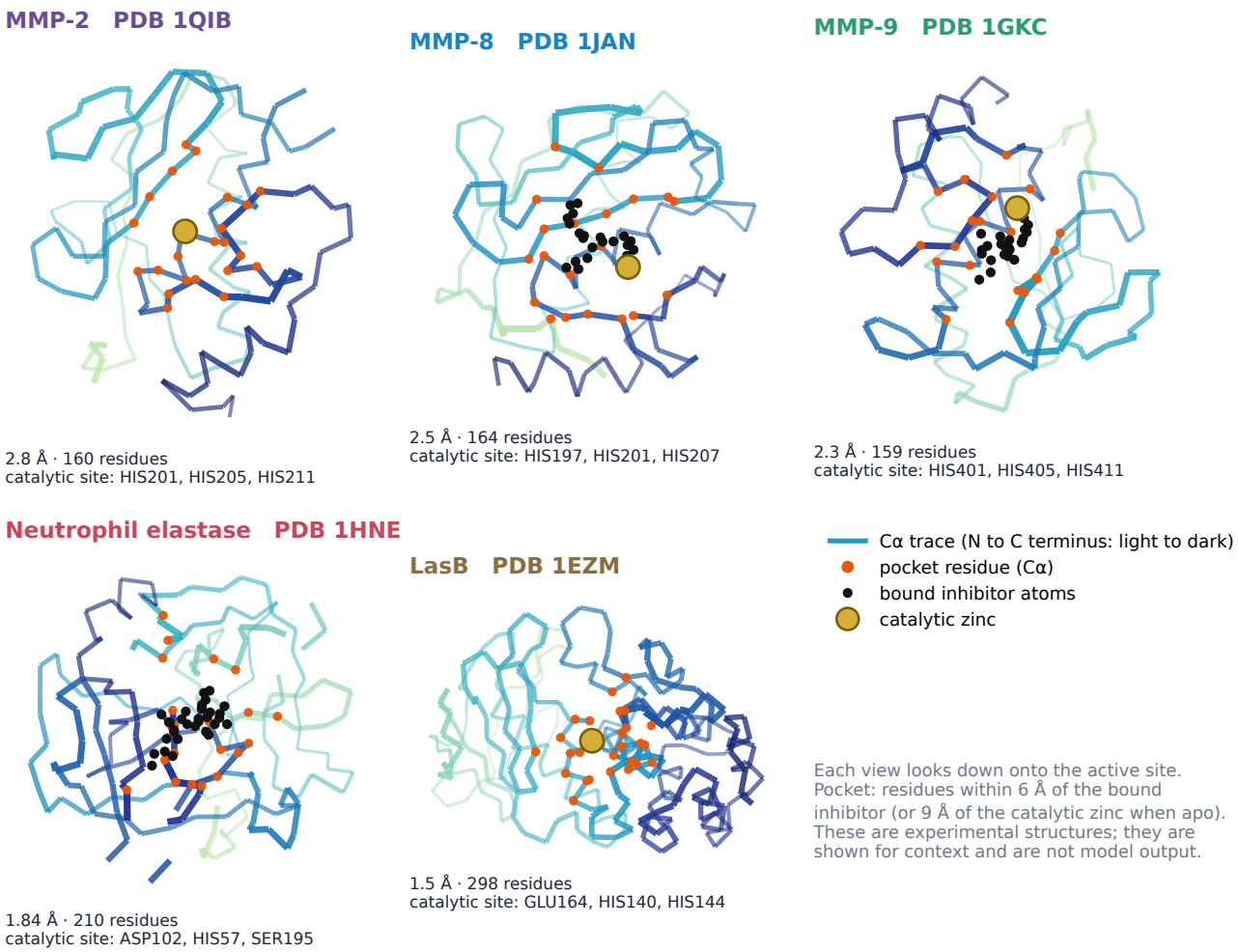


Figure 16: Cα traces of the five structures, each oriented toward the active site (depth-cued; N to C terminus from light to dark). Orange: pocket residues; gold: catalytic zinc; black: bound inhibitor atoms. The structures are experimental and are shown for context; they are not model output.

Held-out measured kinetics

Kridel et al. [1] measured k_{cat}/K_m for MMP-9 on eleven synthetic peptides and detected no activity on a twelfth (C11). Training windows that coincide with a validation peptide on at least seven positions (four windows) were removed from every training set and the MMP-9 model was retrained before scoring. Each peptide was scored as a gapped window and rank-correlated with $\log_{10} k_{\mathrm{cat}}/K_m$.

Peptide	Window (P4–P4′)	Positions	MMP-9 k_{cat}/K_m ($\mathrm{M^{-1}s^{-1}}$)	LR logit	PSSM
C15	`GPRQITA-`	7	188,000	+2.07	+5.41
A11	`IPRTLTA-`	7	67,500	+1.30	+3.67
A6	`GPRAVSTT`	8	61,000	+1.24	+4.20
A11m1	`IPRRLTA-`	7	160,000	+2.43	+5.66
A11m2	`IPRTATA-`	7	2,000	+1.15	+2.35
A10	`GPLFYSVT`	8	25,600	+0.33	+2.70
B1	`QPHYLTTA`	8	8,800	+0.81	+3.10
A18	`RPAKSTA-`	7	5,800	+1.28	+4.96
A7	`-SGLKAL`	6	12,400	+0.77	+3.08
A19	`GSRYLTA-`	7	13,200	+0.74	+3.02
A16	`SGRRLLSR`	8	1,200	+0.68	+0.78
C11	`SGRRLIHH`	8	not detected	+0.30	+2.47

The table lists the validation peptides as P4–P4′ windows with measured k_{cat}/K_m and the scores of the retrained model. The next table gives the rank association; the permutation p-value is two-sided, intervals are bootstrap intervals over peptides, and the last rows rank C11 below every measured peptide.

Score	Set	Spearman ρ	95% bootstrap CI	Permutation p
Logistic regression	n = 11 measured	0.58	[-0.23, 0.93]	0.068
Logistic regression	n = 12 (C11 ranked lowest)	0.68	[0.02, 0.96]	0.017
PSSM	n = 11 measured	0.70	[-0.00, 0.96]	0.020
PSSM	n = 12 (C11 ranked lowest)	0.75	[0.18, 0.93]	0.006
Standardised mean	n = 11 measured	0.57	[-0.21, 0.90]	0.074
Standardised mean	n = 12 (C11 ranked lowest)	0.66	[0.06, 0.91]	0.021

The association is positive for every scorer and set. For the PSSM, $\rho = 0.70$ on the eleven measured peptides (permutation $p = 0.020$; 95% interval $[-0.00, 0.96]$); for the logistic regression $\rho = 0.58$ ($p = 0.068$). The intervals are wide and, for the logistic regression on $n = 11$, include zero. With $n = 11$ and a set that was itself selected for high activity, this is evidence of a positive association and not of accurate rate prediction, and the six tests are not corrected for multiplicity. By the logistic regression the three highest-scored peptides (A11m1, C15, A11) are the three most active; its largest miss is A10 (25,600 $\text{M}^{-1}\text{s}^{-1}$, the fifth most active), which it scores second-lowest. The PSSM scores the least active measured peptide (A16) lowest but ranks A18 (5,800) third, its largest miss.

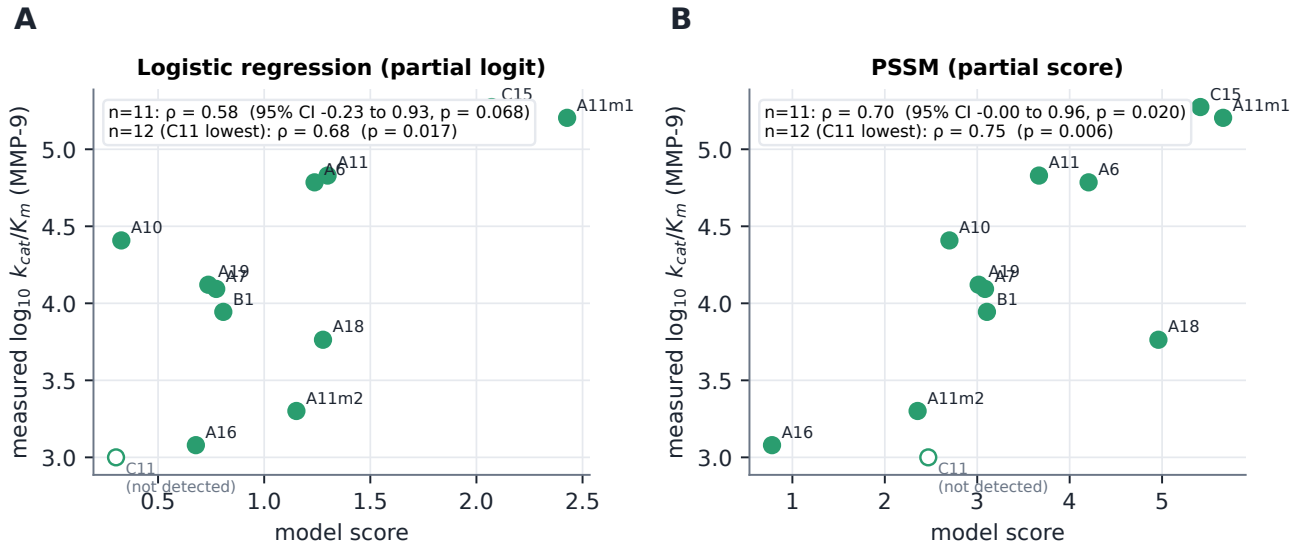


Figure 17: Model score against measured MMP-9 $\log_{10} k_{\text{cat}}/K_m$. Open marker: C11, no activity detected (plotted at the floor of the axis).

Matched point mutants

Two peptides are single-substitution mutants of A11 (SGKIPRT↓LTA): A11m1 replaces Thr with Arg at P1; A11m2 replaces Leu with Ala at P1′. Because the background is identical, each isolates one substitution.

Pair	Substitution	Measured fold change ($\log_2$)	LR Δ (bits)	PSSM Δ (bits)
A11→A11m1	T7R at P1 (Thr to Arg)	2.37× (+1.25)	+1.63	+1.99
A11→A11m2	L8A at P1’ (Leu to Ala)	0.03× (-5.08)	-0.21	-1.31

The model predicts the correct direction for both and approximately the correct size for the first (+1.63 bits from the logistic regression and +1.99 from the PSSM, against a measured +1.25 $\log_2$ units). For the second it predicts a decrease but underestimates it: −0.21 bits (logistic regression) and −1.31 bits (PSSM) against a measured −5.08. Large kinetic effects of a single key residue are compressed by a model trained to distinguish cleaved from uncleaved sites, not fast from slow *(confirmed)*.

VII. Part F: Application to Sensor Candidate Design

Overview

The candidate design analysis defines sensor cores as short motifs and scores them as eight-residue windows. Three of the four cores, PRSLSGLK, PLGLAGQS and PLGVALAA, are motifs whose scissile bond lies after the third residue (Pro-Arg-Ser↓Leu and Pro-Leu-Gly↓Leu, with P3 = Pro and P1′ = Leu, matching the consensus of Kridel et al. [1]). Read as a P4–P4′ window, whose bond follows the fourth character, every residue is displaced by one position from its intended subsite: for PRSLSGLK the window has P2 = S and P1′ = S in place of P2 = R and P1′ = L. The design plan itself states the intended features as P2 = Arg, P1′ = Leu, P2′ = Ser, which the displaced window does not encode.

Assumptions

- **The scissile bond of each core lies where its motif places it.** After the third residue for the Pro-X-X↓Leu cores and after the fourth for AAPV↓SGIG *(unconfirmed).*
- **Device constraints are those of the design filter.** No Cys (reserved for the gold-thiol anchor), no Trp (electroactive near the methylene blue window), net charge −1 to +2, no Pro-Pro, core mass 600–1100 Da. These are team design rules, not model output *(confirmed).*
- **Optimised sequences extrapolate.** Candidates selected from a random pool lie far from any training window, where neither the models nor their percentile scales have been checked *(unconfirmed).*

Equations

Each candidate carries the position after which its bond lies and is scored as a gapped window (Part IV): unknown positions (the P4 of the three cores, and any residue beyond P4′) are marginalised as described there, and percentiles use a null with the same positions masked. The specificity margin is Eq. (IV.3).

Results: effect of the alignment correction

The model responded correctly to the displaced input: PRSLSGLK scored at the 7th percentile for MMP-9. Scored as the gapped window `-PRSLSGL` with the bond after the third residue, it scores at the 100th percentile, and at the top for MMP-2 and MMP-8 as well. The correction also changes a design conclusion. The 3×3 specificity matrix of the MMP-2/MMP-9/NE design analysis has condition number 61.7 with the displaced windows (poorly conditioned; pivot to a two-plex, by the analysis's own rule) and 16.1 with the corrected windows (marginal). The corrected matrix still shows that the MMP-2 and MMP-9 candidates are not separable by score, as the selectivity results of Part V predict *(confirmed).*

Candidate	Core	Scored window	Target	Margin (3-plex)	Pool percentile
Cand1 MMP9	`PRSLSGLK`	`-PRSLSGL`	MMP-9	-0.00	16
Cand2 MMP2	`PLGLAGQS`	`-PLGLAGQ`	MMP-2	-0.00	0
Cand4 NE	`AAPVSGIG`	`AAPVSGIG`	NE	+0.51	26
Cand3 MMP2b	`PLGVALAA`	`-PLGVALA`	MMP-2	-0.04	0

The table gives the candidates scored in the MMP-2/MMP-9/NE panel with correctly aligned, gapped windows; the pool percentile is that of the margin within the PSSM top-3 enumeration pool for the target, with the same gap pattern.

Results: candidate landscape in the wound panel

The figure and table below rank candidates in the five-protease wound panel (MMP-2, MMP-8, MMP-9, NE, LasB). The pool combines 150,000 random windows with the PSSM top-3 enumerations of each target, filtered by the device constraints.

- **The best-scoring sequences are model optima, not recommendations.** They are extreme points of a search over a random pool and are hypotheses for wet-lab testing.
- **The MMP-family reference cores have margin near zero.** The corrected MMP-9 cores (`IPRTLTA-` and `-PRSLSGL`) score at the top for MMP-2, MMP-8 and MMP-9 alike, so no selectivity among MMPs is predicted. Kridel et al. found A11 to be 2.9-fold and 9.6-fold more active on MMP-9 than on MMP-13 and MMP-7 [1]; neither is in the panel, so this cannot be tested here.
- **The panel changes the answer.** AAPVSGIG has margin +0.51 in the three-protease panel and 0.00 in the wound panel, because the LasB model, trained largely on Gly/Ala/Pro-rich elastin sequence, scores it at the 99th percentile. Whether LasB cleaves AAPV-type sites cannot be settled from these data *(unconfirmed).*

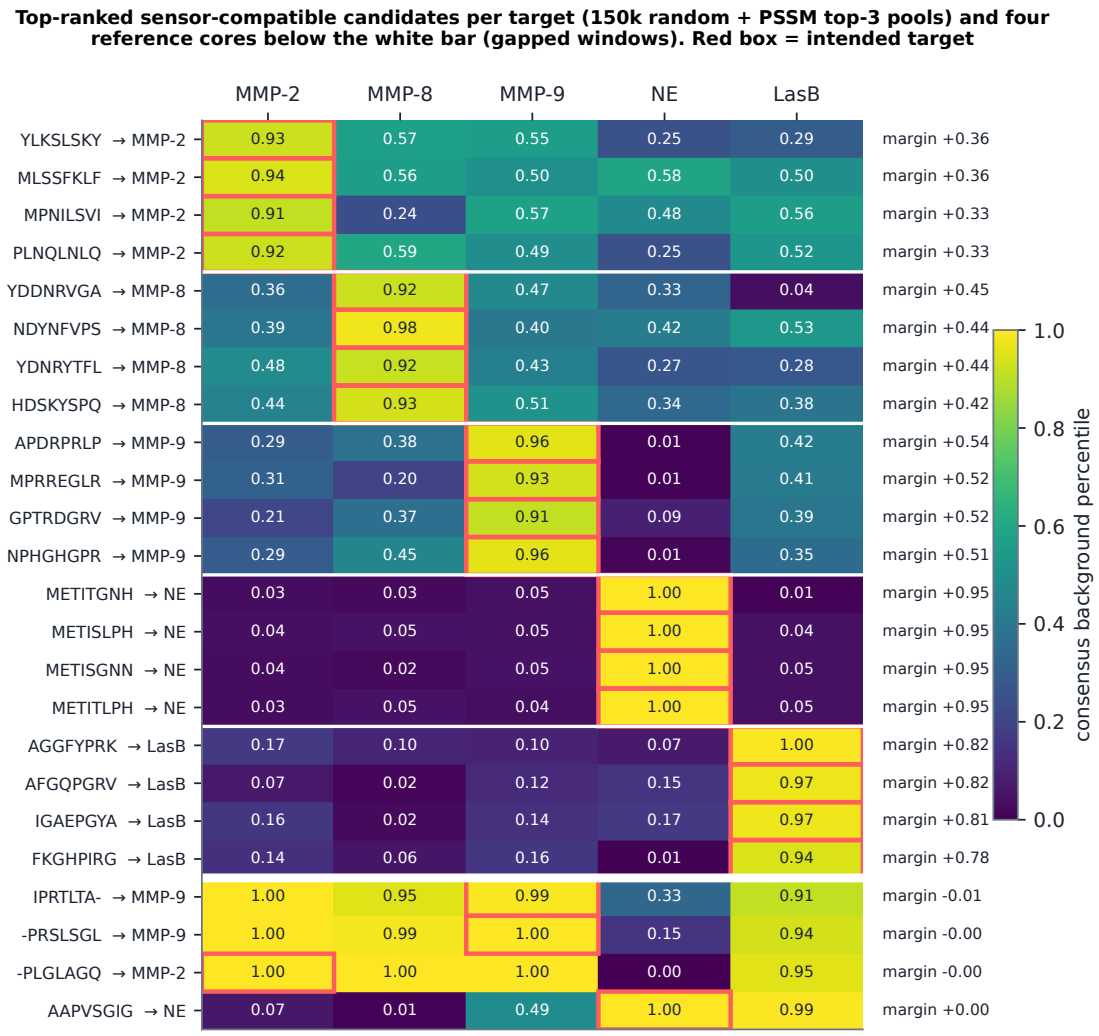


Figure 18: Consensus background percentiles of the four highest-margin sensor-compatible candidates for each target (above the white bar) and of four reference cores (below), across the five wound-panel proteases. Red boxes mark the intended target; margins are listed at the right. Reference cores are scored as gapped windows.

Window	Target	MMP-2	MMP-8	MMP-9	NE	LasB	Margin
YLKSLSKY	MMP-2	0.93	0.57	0.55	0.25	0.29	+0.36
MLSSFKLF	MMP-2	0.94	0.56	0.50	0.58	0.50	+0.36
MPNILSVI	MMP-2	0.91	0.24	0.57	0.48	0.56	+0.33
PLNQLNLQ	MMP-2	0.92	0.59	0.49	0.25	0.52	+0.33
YDDNRVGA	MMP-8	0.36	0.92	0.47	0.33	0.04	+0.45
NDYNFVPS	MMP-8	0.39	0.98	0.40	0.42	0.53	+0.44
YDNRYTFL	MMP-8	0.48	0.92	0.43	0.27	0.28	+0.44
HDSKYSPQ	MMP-8	0.44	0.93	0.51	0.34	0.38	+0.42
APDRPRLP	MMP-9	0.29	0.38	0.96	0.01	0.42	+0.54
MPRREGLR	MMP-9	0.31	0.20	0.93	0.01	0.41	+0.52
GPTRDGRV	MMP-9	0.21	0.37	0.91	0.09	0.39	+0.52
NPHGHGPR	MMP-9	0.29	0.45	0.96	0.01	0.35	+0.51
METITGNH	Neutrophil elastase	0.03	0.03	0.05	1.00	0.01	+0.95
METISLPH	Neutrophil elastase	0.04	0.05	0.05	1.00	0.04	+0.95
METISGNN	Neutrophil elastase	0.04	0.02	0.05	1.00	0.05	+0.95
METITLPH	Neutrophil elastase	0.03	0.05	0.04	1.00	0.05	+0.95
AGGFYPRK	LasB	0.17	0.10	0.10	0.07	1.00	+0.82
AFGQPGRV	LasB	0.07	0.02	0.12	0.15	0.97	+0.82
IGAEPGYA	LasB	0.16	0.02	0.14	0.17	0.97	+0.81
FKGHPIRG	LasB	0.14	0.06	0.16	0.01	0.94	+0.78
IPRTLTA-	MMP-9	1.00	0.95	0.99	0.33	0.91	-0.01
-PRSLSGL	MMP-9	1.00	0.99	1.00	0.15	0.94	-0.00
-PLGLAGQ	MMP-2	1.00	1.00	1.00	0.00	0.95	-0.00
AAPVSGIG	Neutrophil elastase	0.07	0.01	0.49	1.00	0.99	+0.00

Results: contribution of each position

The scan below scores every single-residue substitution of IPRTLTA- (the A11 core). Panel A shows the change in the MMP-9 logistic-regression log-odds, which does not saturate; panel B shows the change in specificity margin, which is near zero for most substitutions because the wild type already lies at the top of the MMP percentile scale. Substitutions at P3 are the most damaging (replacement of Pro costs roughly 1 to 2.5 log-odds units); Thr to Arg or Lys at P1 raises the score, consistent with the measured effect of A11m1; and P1′ is comparatively tolerant in this model, the same underestimation seen for A11m2.

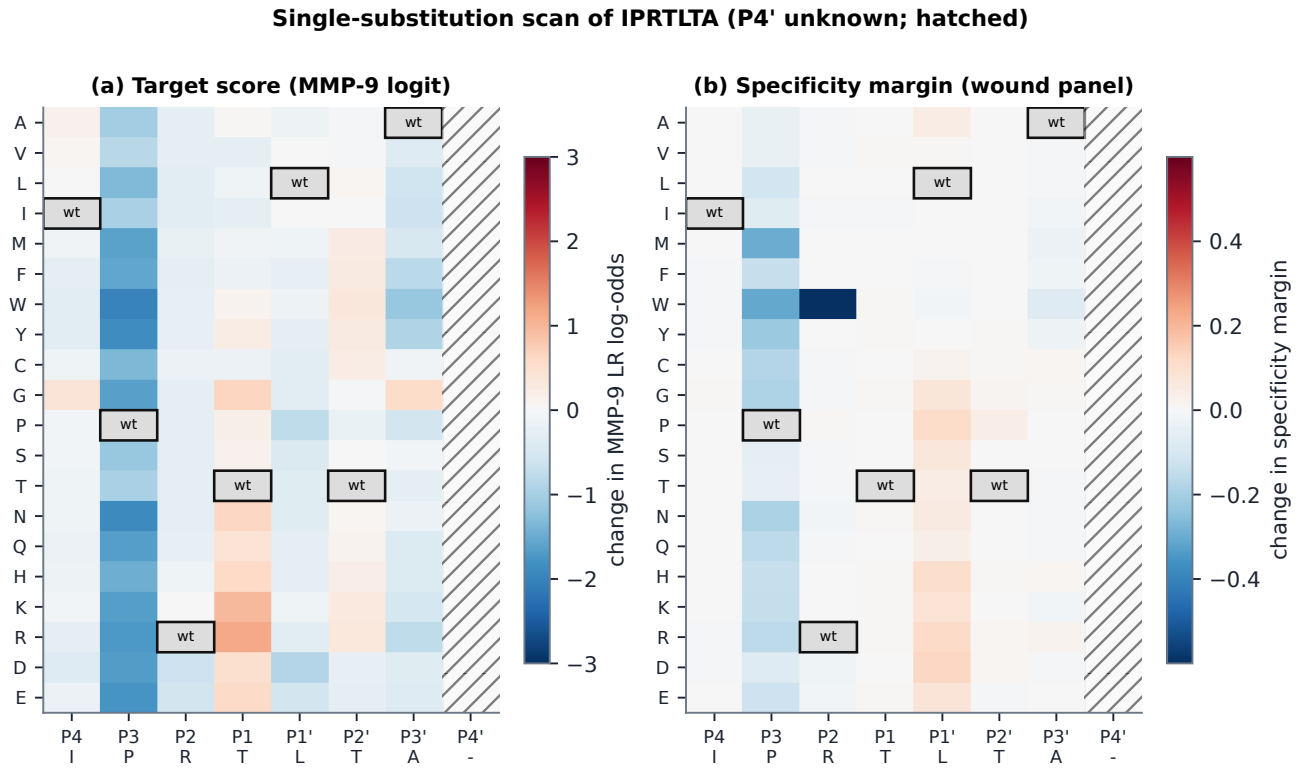


Figure 19: Single-substitution scan of the A11 core. **A:** change in MMP-9 logistic-regression log-odds. **B:** change in specificity margin in the wound panel.

Result: the interactive tool

The browser tool embeds the seven models and the five structures and loads the 3Dmol.js renderer from a CDN. Its JavaScript scoring engine was compared with the Python package on 154 random and 8 gapped windows across all seven proteases: logistic-regression scores, PSSM scores, per-position contributions and percentiles of full windows agree to about 10^{-15}, and percentiles of gapped windows agree within 0.012, the only difference being the random-number generator behind the matched null *(confirmed)*. The built page was exercised in a headless browser (no script errors; controls, presets, scan clicks and structure switching respond; no horizontal overflow at phone width) *(confirmed)*.

VIII. Summary: Confirmed Results, Working Assumptions, and Open Questions

Part	Component	Status	
A	Positive windows	Rebuilt from MEROPS, Eckhard and Yang sources; counts reproduce the earlier pipeline; 75 of 75 elastin rows verified against UniProt	*(confirmed)*
A	Removal of synthetic LasB positives	Rule-generated and hand-typed windows removed; LasB uses source windows only	*(confirmed)*
A	Negatives as true non-substrates	Unlabelled windows assumed negative; contamination bounded but unknown	*(unconfirmed)*
A	LasB positives	76% from one protein (elastin); model behaves as an elastin-likeness score	*(confirmed)*
B	VHSE table and labels	Regenerated from the archived source and unit-tested; the earlier table had 35 of 160 values wrong	*(confirmed)*
B	Additive position-specific model	Trees do not exceed additive models by more than 0.007 AUROC	*(confirmed)*
B	PSSM smoothing β, ϵ	Fixed, not tuned	*(unconfirmed)*
C	Percentile comparability	Uniform under the null by construction; resolution 10^{-3}	*(confirmed)*
C	Harmonic-mean consensus and max-competitor margin	Reasoned, not validated against a selectivity ground truth	*(unconfirmed)*
D	Detection performance	Cluster-aware AUROC 0.77 (MMP-9) to 0.95 (faecal); random and clustered folds differ by at most 0.007; label-shuffle 0.49–0.53	*(confirmed)*
D	Representation choice	VHSE leads below about 200 positives; PSSM and one-hot lead on MMP-2 and MMP-8; consensus retained	*(confirmed)*
D	Panel negatives	Improve cross-protease AUROC on every protease at 0.01–0.02 cost against background	*(confirmed)*
D	Selectivity within the MMPs	Held-out AUROC 0.53–0.72; partly reflects genuine promiscuity in the labels	*(confirmed)*
E	MMP-9 consensus recovery	Pro at P3 first, Leu/Ile at P1′ first and second; Ser/Thr not recovered, as expected from the phage-vector artefact	*(confirmed)*
E	Structures	Catalytic-site residues derived from coordinates; HEXXHXXGXXH confirmed in three MMPs	*(confirmed)*
E	Measured kinetics ($n = 11$)	Positive rank association (PSSM $\rho = 0.70$, logistic regression 0.58); intervals wide; not corrected for multiplicity	*(confirmed)*
E	Single-mutant effects	Direction correct for both; Leu to Ala at P1′ underestimated	*(confirmed)*
F	Alignment of sensor cores	Three of four cores corrected to scissile-aware windows; 3-plex condition number 61.7 → 16.1	*(confirmed)*
F	Optimised candidates	Extrapolate beyond the training distribution	*(unconfirmed)*
F	Prospective wet-lab test	Not yet performed	*(unconfirmed)*

Bottom line. The models discriminate confirmed cleavage windows from background reliably (cluster-aware AUROC 0.77 to 0.95), the result is robust to near-duplicate leakage and to label shuffling, and the learned specificity agrees with published MMP-9 specificity and with the active-site structures of the enzymes. What the models cannot do is equally clear. Selectivity among the MMPs cannot be obtained from a P4–P4′ window, the LasB model reflects its single-protein source, scores rank resemblance to observed sites and are not rates, and the external validation ($n = 11$) supports direction but not magnitude. What requires direct experimental attention is a prospective test: measured relative activities of new peptides across the panel, against which the percentile margin can be checked.

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